Temporal Patterns and Pixel Precision: Satellite-Based Crop Classification Using Deep Learning and Machine Learning

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Declarations

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Abstract

Accurate crop classification from satellite imagery is essential for agricultural monitoring, yield forecasting, and food security assessment, yet remains particularly challenging in dryland farming regions of the Global South where spectral similarity between crop types, irregular phenological cycles, and persistent cloud cover complicate classification. Most crop-classification studies report performance under random or field-wise cross-validation *within* a single region, which measures in-distribution fit rather than the spatial transfer that operational mapping actually requires. Using a labeled field-boundary dataset from the Western Cape of South Africa, we benchmark nineteen classical, deep learning, and patch-based models twice: under conventional in-region field-wise validation, and on a spatially disjoint holdout tile. The two protocols produce sharply different conclusions. Under in-region validation the dense temporal deep nets rank at the top (macro-F1 0.72–0.78); on the spatial holdout they collapse, and the CNN-BiLSTM ensemble — the model conventional field-level evaluation singles out as best — suffers the largest degradation of any model (macro-F1 0.72 to 0.46, a 0.26 drop), falling to near-worst. The ranking inverts: a TabNet ensemble becomes the best single model (macro-F1 0.60, Kappa 0.54), and simple classical baselines — gradient-boosted trees and even logistic regression (macro-F1 0.56) — transfer with the smallest gaps. We argue this is not incidental but mechanistic: the sparse, axis-aligned feature selection that made Random Forest and gradient-boosted trees the remote-sensing workhorse for two decades is precisely the inductive bias that governs spatial transfer, and TabNet inherits it (attentive sparsemax feature masks) while dense CNN/RNN/attention temporal nets discard it and overfit region-specific signal. A complementary lever — field-level aggregation as variance reduction — recovers transfer for dense models (L-TAE improves out-of-sample when aggregated) but adds nothing for already-sparse models. We further show, via a training-set reduction experiment, that the most transferable models retain near-full holdout accuracy with as little as 25% of the training fields. Classical models — including XGBoost, Random Forest, and LightGBM — were paired with automated time-series feature extraction via `xr_fresh`, evaluated here for the first time in crop classification. The central message for the remote-sensing community is that model selection in crop mapping is an artifact of the validation protocol: spatial generalization, not in-distribution accuracy, must drive it.

Keywords: Crop classification, Satellite imagery, Deep learning, Machine learning, Remote sensing

1 Introduction

Accurate and timely crop classification is foundational to precision agriculture, enabling yield forecasting, resource optimization, food security assessment, and climate-resilient decision-making [47, 1]. The widespread availability of high-resolution Earth observation data, especially from the Sentinel-2 satellite constellation, has transformed the landscape of agricultural monitoring. With 13 spectral bands, spatial resolutions ranging from 10 to 60 meters, and a revisit frequency of approximately five days, Sentinel-2 offers dense multi-temporal imagery rich in spatial and spectral information.

Despite these advancements, reliable crop classification remains a complex task on plots throughout Africa and the Global South. Spectral similarities between crops, temporal variability in planting and phenological cycles, and atmospheric disturbances such as cloud cover hinder classification accuracy. In Africa, this is compounded by the complexities of rain-fed dryland farming [47]. Traditional machine learning methods often rely heavily on manually engineered features, such as vegetation indices or statistical summaries, which may capture pertinent temporal and spectral dynamics but are often more limited in complexity of features. A number of methods like XGBoost (XGB) and Random Forest (RF) have proven effective across crop classification and broader agricultural prediction tasks such as yield and loss estimation [34, 8, 35, 36]. However in regions with significant differences in cropping patterns and practices, the complexity of the task can outstrip the exploitable variability in the training data [37, 34]. Newly released automated feature extraction methods like that implemented in xr_fresh offer the possibility of rich feature sets for use in machine learning but it remains untested in the literature [37]. Despite their effectiveness, machine learning methods can be inconsistent [24]. Several factors contribute to this variability, including hyperparameter selection [9], sample quality and quantity [10], selection of training and testing splits [12], imbalanced classes [12], and feature generation and selection [11].

To address these limitations, deep learning approaches have gained traction in crop type mapping, with both Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) emerging as particularly promising architectures. RNNs are well suited to the sequential nature of remote sensing time series, as their internal hidden states allow them to retain a form of memory across time steps, capturing the temporal dynamics of crop phenology that static or summary-based features often miss. This capacity to model temporal structure has proven especially valuable for crop classification in data-rich environments, where the full growing season trajectory of spectral signals can be exploited [38, 25]. The lack of high quality training data in the Global South and lack of reliable cloud-free imagery has pushed many of the most successful efforts into the realm of precision agriculture such as applications for hyperspectral and drone-derived imagery [34]. Nevertheless, recent work applying deep learning to agricultural mapping in developing world contexts has demonstrated encouraging results. Ayushi et al. [40] applied a U-Net fully convolutional encoder-decoder architecture to a crop type mapping task in Kenya, achieving a macro F1 of roughly 0.74, while Nowakowski et al. [46] fine-tuned pre-trained convolutional networks via transfer learning for crop-type mapping in Malawi, reaching around 83% accuracy. Rustowicz et al. [32] released the first crop-type semantic-segmentation dataset for smallholder farms in Ghana and South Sudan, underscoring the difficulty of small fields, frequent cloud cover, and scarce labels in such settings. However questions remain about the generalizability of deep learning models particularly in complex environments with limited training data [42].

Hybrid approaches offer a compelling opportunity by combining the representational power of deep learning with the interpretability and efficiency of traditional machine learning, and several studies have demonstrated their value in agricultural remote sensing contexts. A common strategy in precision agriculture involves using a pre-trained deep learning model as a feature extractor especially, with the resulting high-level representations passed to a traditional classifier for the final prediction. Espejo-Garcia et al. [43] fine-tuned architectures including MobileNet, DenseNet, and Xception before replacing their fully connected layers with shallow classifiers such as Support Vector Machines, gradient boosting, and logistic regression for weed identification, while Koklu et al. [44] found that, on a single grapevine-leaf dataset, pairing MobileNetV2 feature extraction with an SVM improved on end-to-end deep learning. Yang et al. [45] more relevant to this study, combined a CNN with a Random Forest classifier for Sentinel-2 crop type mapping, leveraging the spatial-spectral learning capacity of the CNN alongside the robustness and resistance to overfitting characteristic of ensemble tree methods. Taken together, these hybrid approaches suggest a productive middle path between the interpretability and data efficiency of classical machine learning and the feature-learning capacity of deep neural networks, and they form part of the methodological landscape against which the approach taken in this study is positioned.

This study addresses a critical yet underexplored gap in the crop classification literature: the systematic comparison of machine learning, deep learning, and hybrid approaches under the constraints typical of the Global South, but with the advantage of an unusually rich, high-quality training dataset. While much of the methodological literature has been developed and validated in data-rich temperate agricultural settings, the Western Cape of South Africa presents a distinct set of challenges — smaller field sizes, rain-fed dryland farming systems, heterogeneous and spatially variable planting rotations, persistent cloud cover during the critical winter growing season, and phenological cycles that do not conform to the assumptions embedded in many widely used models. We leverage this rare combination of environmental complexity and labeled data richness to conduct a rigorous evaluation of methods that are seldom compared within a single study and region. Our experiments utilize a labeled field-boundary dataset provided by Radiant Earth's MLHub [2], capturing ground-truth crop types for agricultural fields in South Africa during the 2017 growing season.

We evaluate multiple deep learning architectures — including CNN-BiLSTM hybrids, TabNet ensembles, light-weight temporal-attention (L-TAE) and temporal-convolution (TempCNN) networks, and 3D CNN models — that extract dynamic phenological features across time, and benchmark these against traditional ensemble methods including XGBoost, LightGBM, and Random Forest. Critically, we evaluate every model twice: under the conventional protocol — random or field-wise cross-validation *within* the training region — and on a spatially disjoint holdout tile that no training pixel touches. This dual evaluation exposes a large and architecture-dependent generalization gap that single-region validation hides entirely, and it reorders the models. Throughout, we restrict our inputs to multi-temporal Sentinel-2 optical imagery alone, without recourse to SAR data, in order to isolate the signal available from spectral time series and maximize applicability to contexts where only optical data is available. All code is publicly available to support reproducibility and future research [3].

This study makes four contributions. First, we show that the sparse, axis-aligned feature-selection inductive bias behind the two-decade dominance of Random Forest and gradient-boosted trees in remote sensing is precisely the property that governs *spatial* transfer, and that TabNet inherits it through attentive sparsemax feature masks while dense CNN, RNN, and attention temporal networks discard it and overfit region-specific signal. Second, we quantify the in-region to spatial-holdout ranking inversion across nineteen models: the dense temporal networks that lead in-region — CNN-BiLSTM in particular, the model conventional evaluation singles out as best — become near-worst on the holdout, while parsimonious models transfer with the smallest gaps. Third, we identify field-level aggregation as a second, substitutable robustness lever, a form of variance reduction that recovers transfer for dense temporal models but is redundant for already-sparse ones. Finally, we characterize data efficiency through a training-set reduction experiment and provide the first evaluation of `xr_fresh` automated time-series feature extraction in crop classification.

This leads to the central research question: across a range of machine learning, deep learning, and hybrid architectures, which approach most reliably classifies crop types from multi-temporal Sentinel-2 optical imagery in a complex dryland farming environment *when evaluated on a spatially disjoint region*, and what properties of a model determine whether its in-distribution accuracy survives the transfer to unseen terrain?

1.1 Related Work

Crop classification using satellite imagery has become a core task in precision agriculture because it supports large-scale crop monitoring, yield estimation, resource management, and food-security analysis. The increasing availability of high-resolution, multi-temporal Earth observation data, particularly from Sentinel-2, has enabled substantial progress in crop mapping, but several challenges remain. These include strong spectral similarity among crop types, temporal variability in planting and phenological cycles, class imbalance, and the difficulty of building methods that generalize across regions and farming systems.

Early work in crop classification relied primarily on classical machine learning methods such as Random Forest, Support Vector Machines, Logistic Regression, and gradient-boosted tree models. These approaches typically operate on hand-crafted inputs derived from spectral bands, vegetation indices, temporal summaries, or other engineered descriptors, and they have often proven effective because they are computationally efficient, interpretable, and robust on structured remote-sensing data [24, 1]. However, their performance depends heavily on feature engineering and domain knowledge, which can limit transferability across crop systems and agroecological regions. This challenge becomes especially pronounced in heterogeneous agricultural landscapes, where differences in management practices, field conditions, and phenological timing can reduce the separability captured by manually designed features [24, 1]. Comparative studies have directly weighed classical machine learning against deep learning for crop identification, albeit largely on close-range crop imagery rather than satellite scenes [13].

These limitations have motivated growing interest in deep learning approaches that learn hierarchical feature representations directly from raw or minimally processed observations. Convolutional neural networks (CNNs) are effective at extracting local spectral and spatial patterns [26], while recurrent architectures such as LSTMs and BiLSTMs are well suited to modeling temporal crop-development signals over a season. More broadly, deep learning has become increasingly important in remote sensing because it reduces reliance on manually constructed predictors and can capture complex temporal structure that summary statistics often miss [25, 38]. Prior work has shown that temporal modeling is especially valuable when crop types are difficult to distinguish from single-date imagery but follow different seasonal trajectories [15, 18].

A particularly important direction in the literature has been the development of hybrid architectures that combine convolutional feature extraction with temporal sequence modeling. For example, Fan et al. proposed a hierarchical convolutional recurrent neural network for Sentinel-2 image classification, showing that combining CNN-based feature extraction with recurrent modeling can improve in-region classification accuracy relative to both classical and standalone deep learning baselines [5]. Similarly, Tufail et al. demonstrated that hybrid CNN-RNN models applied to multi-temporal Sentinel-2 imagery and red-edge vegetation indices can achieve very strong crop-mapping performance by jointly learning spatial context and temporal phenology [6]. Bandar and Coşkunçay likewise showed that combining an attention-based BiLSTM with a temporal CNN improves crop classification from Sentinel-2 time-series data, reinforcing the broader value of spatiotemporal representation learning in agricultural remote sensing [7].

Beyond fully end-to-end deep architectures, several studies have explored hybrid pipelines that combine learned representations with classical machine learning classifiers. Espejo-Garcia et al. used deep convolutional backbones as feature extractors before applying conventional classifiers for weed identification, while Koklu et al. found that deep features combined with an SVM improved on end-to-end models on a grapevine-leaf dataset [43, 44]. In a setting more directly related to this study, Yang et al. combined CNN-based representation learning with a Random Forest classifier for multi-temporal crop classification from Sentinel-2 imagery [45]. Taken together, these studies suggest that hybrid approaches can provide a useful middle ground between the flexibility of deep learning and the robustness and interpretability of classical models.

The representation of input data is another central theme in the literature. Sentinel-2 has become a preferred platform for crop monitoring because of its high revisit rate and multispectral richness, and these data are exploited either through engineered descriptors (spectral indices, temporal summaries) fed to classical learners or through end-to-end models that learn directly from temporal reflectance sequences. A recurring assumption is that richer, learned representations should generalize better than compact engineered ones; whether this holds under genuine spatial transfer — as opposed to within-region validation — is less well established and is a central question of this study.

A second, and for our purposes critical, gap concerns *how generalization is measured*. Much of the crop-classification literature reports accuracy under random or field-wise cross-validation drawn from a single region or scene [12]. Such protocols hold out individual pixels or fields but not *space*: training and test samples share the same atmospheric conditions, soils, management calendars, and sensor-acquisition geometry, so the reported numbers reflect in-distribution fit rather than transfer to new terrain. Because operational crop maps are produced precisely by applying a model to regions where no labels were collected, this distinction matters, yet studies that re-rank models under a spatially disjoint holdout remain rare [42]. A handful of architectures explicitly target cross-scenario generalization—Cropformer, for example, couples convolutional and transformer features with self-supervised pre-training to handle full-season, in-season, few-sample, and transfer settings, including a cross-region evaluation [33]—though such cross-scenario architectures remain the exception, and are rarely benchmarked against classical and deep baselines on a common spatially disjoint holdout. This is compounded in the Global South, where rain-fed dryland farming, heterogeneous field conditions, irregular planting calendars, and persistent cloud cover make spatial transfer especially demanding [47, 34].

Our study is positioned within this gap. Rather than evaluating a single architecture in isolation, we conduct a systematic comparison of classical machine learning, deep learning, and patch-based approaches under *both* in-region field-wise validation and a spatially disjoint holdout, in the Western Cape of South Africa. We build on prior work in temporal modeling, hybrid learning, and ensemble-based classification, while also evaluating `xr_fresh` as an automated time-series feature-extraction framework in a crop-classification setting [37]. This allows us to ask not only which approach is most accurate, but which model properties cause in-distribution accuracy to survive — or fail to survive — the move to unseen terrain.

2 Data Description

The dataset used for this study consists of two primary components: satellite imagery and field boundary data, as sourced from [16].

2.1 Study Area and Description

The study was conducted in an agriculturally productive region of the Western Cape of South Africa, characterized by diverse agroecological conditions and cropping systems. Major crop types in the dataset include wheat, barley, canola, small grain grazing, and lucerne/medics (alfalfa). The region spans varying climatic zones, from semi-arid to subtropical, and exhibits seasonal differences in rainfall and temperature, contributing to distinct crop phenologies.

The labeled data are organized into Sentinel-2 UTM grid tiles (zone 34S). Two adjacent tiles (`34S_19E_258N` and `34S_19E_259N`) form the training region, comprising 4,150 labeled fields covering 622 km^2 of cultivated area (mean field size 15.0 ha, median 11.0 ha). A third tile (`34S_20E_259N`), immediately east of and adjacent to the training tiles, is reserved as a holdout region for out-of-sample evaluation and contains 2,417 labeled fields over 282 km^2 (mean 11.7 ha, median 6.9 ha). The tiles do not overlap, so no pixel or field is shared between training and holdout, and transfer to the holdout must contend with different soils, field geometries, and acquisition conditions. Because the holdout nonetheless lies within the same agroecological zone and growing-season calendar as the training tiles, this is a relatively *local* spatial shift; the generalization gaps we report should accordingly be read as a conservative lower bound on the degradation expected under transfer across regions or climates (Section 5). Unless otherwise noted, "in-region" results refer to a field-wise (`fid`-stratified) train/validation/test split within the training region, while "out-of-sample" or "holdout" results refer to predictions on the disjoint tile.

Field parcels differ in size, shape, and management practices, introducing heterogeneity into the dataset. Sentinel-2 imagery was collected monthly across the 2017 growing season, covering key phenological stages from emergence to harvest. Each field carries a single crop-type label for the season. This temporal resolution is vital for capturing crop-specific growth patterns, which are often indistinguishable in single-date imagery.

2.2 Remote Sensing Data

Sentinel-2 Level-2A optical surface-reflectance data were acquired over the study area in South Africa for the January–December 2017 growing season. Individual scenes were cloud-masked using a 30% `s2cloudless` cloud-probability filter combined with a morphological cloud-shadow mask clipped to field boundaries. Six features were produced for each month: four optical bands — B2 (Blue, 490 nm), B6 (Red Edge 1, 740 nm), B11 (SWIR 1, 1610 nm), and B12 (SWIR 2, 2190 nm) — together with the Enhanced Vegetation Index (EVI) and Hue, a colorimetric descriptor of vegetation condition [8]. The reflectance bands were formed as monthly median composites, whereas EVI and Hue were computed *per scene* and then composited to a monthly value (detailed below). All layers were scaled to integer reflectance ($\times$10,000) and exported as 10 m GeoTIFFs via Google Earth Engine. This band subset was chosen to span the spectral regions most informative for separating the target crops while limiting redundancy: the blue band (B2) constrains atmospheric and soil-brightness effects and enters the EVI formulation; the red-edge band (B6) is sensitive to chlorophyll content and canopy structure, which differ across cereals and broadleaf crops; and the two SWIR bands (B11, B12) respond to leaf and soil moisture and to senescence dynamics that help discriminate phenologically similar cereals such as wheat and barley (Figure 4). Red-edge bands in particular are repeatedly reported among the most informative Sentinel-2 features for separating spectrally similar crops [6, 34]. We deliberately did not retain the red (B4) and near-infrared (B8) bands as standalone features. Although these are the classical vegetation bands, their information is not discarded: it enters the feature set through EVI — which is itself a function of near-infrared, red, and blue reflectance — and through the chlorophyll-sensitive red-edge band (B6), which is spectrally adjacent to the near-infrared. Prior feature-importance analysis on comparable Sentinel-2 crop data motivated this compact subset as a way to limit redundancy among highly correlated bands [8]. Importantly, EVI and Hue were computed from the complete per-scene reflectance before this subset was retained, so the index values are unaffected by the band selection.

Eleven monthly composites were produced over the growing season; June 2017 yielded no usable imagery and May 2017 was discarded due to persistent cloud cover, leaving 10 valid monthly observations per feature. Each pixel was consequently represented by a 60-dimensional time-series feature vector (six features across ten months), enabling the capture of rich phenological dynamics across the season.

A key methodological point concerns the order of operations for the derived indices. EVI and Hue were computed *per scene* — that is, from the simultaneously acquired bands of each individual Sentinel-2 acquisition — and only then composited, by taking the monthly median over the cloud-free per-scene index values. This is distinct from, and avoids the pitfall of, computing an index from already-composited bands: because each per-scene EVI uses bands observed at the same instant, the per-acquisition band simultaneity that the EVI formulation assumes is preserved, and the resulting monthly EVI is a physically consistent index rather than an artifact of mixing reflectances drawn from different dates. The monthly median then serves only to reduce residual cloud and acquisition noise across the index time series, exactly as it does for the raw reflectance bands.

By relying exclusively on optical data rather than fusing it with SAR imagery [4, 41], this study demonstrates the capability of multi-temporal spectral analysis alone for robust crop classification. This approach is directly transferable to emerging high-resolution optical platforms such as PlanetScope, reinforcing the operational relevance of dense, temporally-aware optical remote sensing for agricultural monitoring at scale.

2.3 Field Boundary Data

The field boundary polygons were drawn from the crop type classification dataset for Western Cape, South Africa [2]. This GeoJSON file delineated each agricultural field as a polygon, with boundaries digitized primarily from 50 cm aerial photography at scales of 1:2,500 for horticultural parcels and 1:7,500 for other fields between May and August 2017. In cases where ground reference was unavailable, Sentinel-2 imagery supplemented the digitizing process. Each polygon fully encompassed the field area and carried a crop-type label obtained via aerial and vehicle surveys conducted from May 2017 to March 2018 [2].

All predictions in our study were made at the field level.

2.4 Crop Types and Distribution

In this study, we classified five crop types: wheat, canola, lucerne/medics (alfalfa), barley, and small-grain grazing.

Figure 1 illustrates the distribution of crops in the study area. The data are notably imbalanced: in the training region lucerne/medics accounts for 1,792 of the 4,150 fields (43%), followed by wheat (753), barley (660), canola (512), and small grain grazing (433); the holdout region is more skewed still, with lucerne/medics comprising 1,360 of 2,417 fields (56%). This imbalance has important implications for model training and evaluation, as classification algorithms may be biased towards the majority class unless appropriate balancing techniques are applied, and it motivates our use of macro-averaged F1 as the primary metric (Section 3.3).

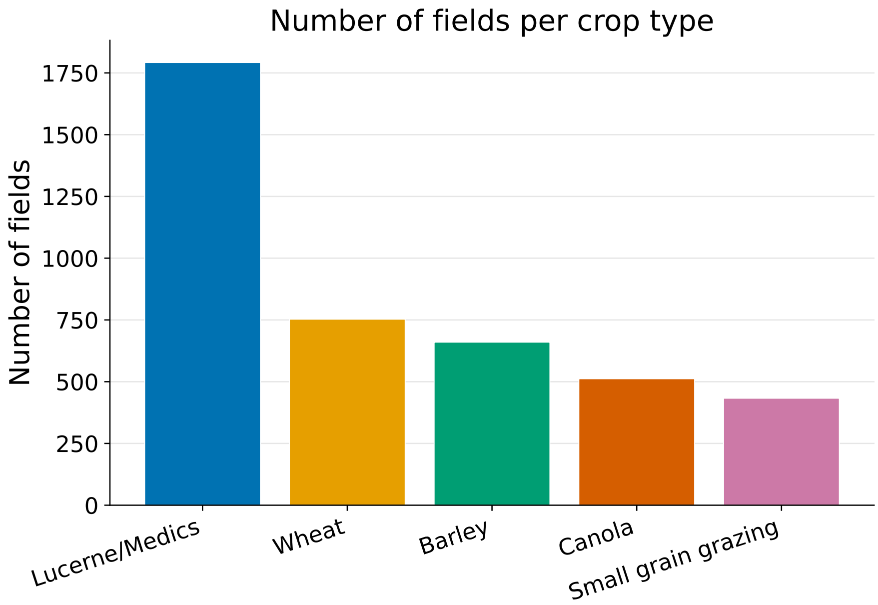

Fig. 1: Number of fields per crop type.

Figure 2 presents the mean field area for each crop type, measured in hectares. Although lucerne/medics represent the majority of fields, their average area (11.6 ha) is the smallest of the five crop types.

Figure 3 displays box plots of spectral band intensities for each crop type, illustrating the distribution and overlap of spectral responses. Although some differences in median and spread are evident, the substantial overlap in spectral intensities across crops demonstrates that classification based solely on a limited set of spectral bands is challenging and prone to confusion. This was particularly true for cereal crops such as wheat and barley, which exhibited highly similar spectral profiles, as shown in Figure 4. The close alignment of their spectral signatures underscores why models often struggle to distinguish between these two crops using spectral data alone [17].

These findings highlight the need to incorporate temporal information that better represents phenological stages throughout the growing season to improve crop classification. The unique temporal dynamics associated with the stages of crop development (e.g., emergence, peak greenness, senescence) provide additional discriminatory power that can help resolve ambiguities between crops with similar spectral signatures [18]. Therefore, integrating both spectral and temporal data was crucial to enhance classification accuracy, especially in complex agricultural landscapes where crops like wheat and barley were spectrally similar.

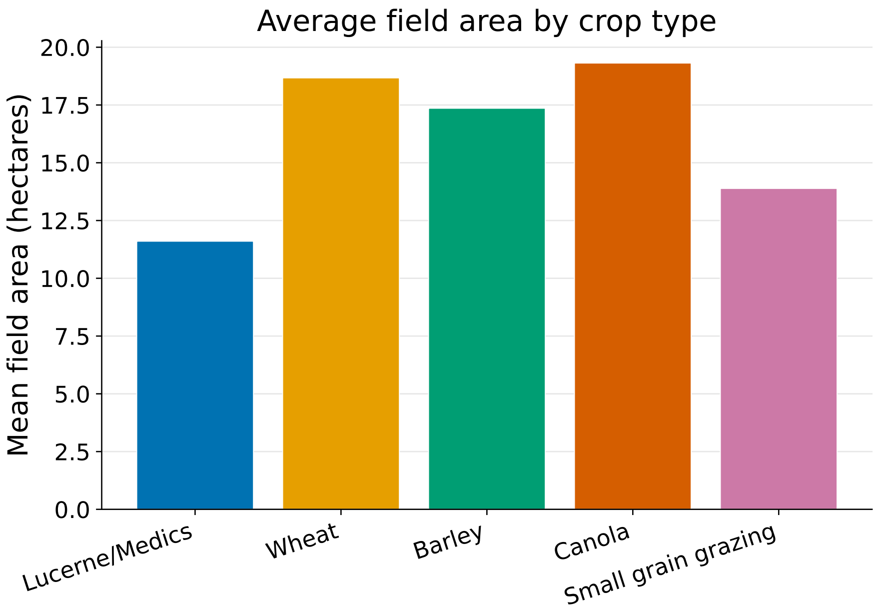

Fig. 2: Average field area by crop type (hectares).

3 Methodology

This study explores three complementary modeling paradigms for crop classification: feature-based machine learning, deep learning, and patch-based approaches. These paradigms differ in how spatial and temporal information from multi-spectral Sentinel-2 imagery is represented and utilized.

Classical machine learning relies on engineered features derived from pixel-level time series, while deep learning models learn hierarchical representations directly from raw data. In contrast, patch-based approaches operate on spatially contiguous regions, enabling the model to capture local spatial structure and contextual relationships that are not available in purely pixel-wise representations. Each paradigm was evaluated at both the pixel level and the field (or patch) level to assess their effectiveness across different representations of the data.

Crop classification from remote sensing data can therefore be approached through three complementary strategies: feature engineering, representation learning, and spatial context modeling.

In the feature engineering paradigm, domain knowledge is used to construct descriptive inputs from raw spectral time series. These include vegetation indices such as NDVI, spectral band ratios, textural measures, and phenological statistics, which are then used as inputs to classical machine learning models such as Random Forests (RF), Support Vector Machines (SVM), and Gradient Boosted Trees (GBT) [1]. In this study, we adopt the `xr_fresh` framework to automatically extract a comprehensive set

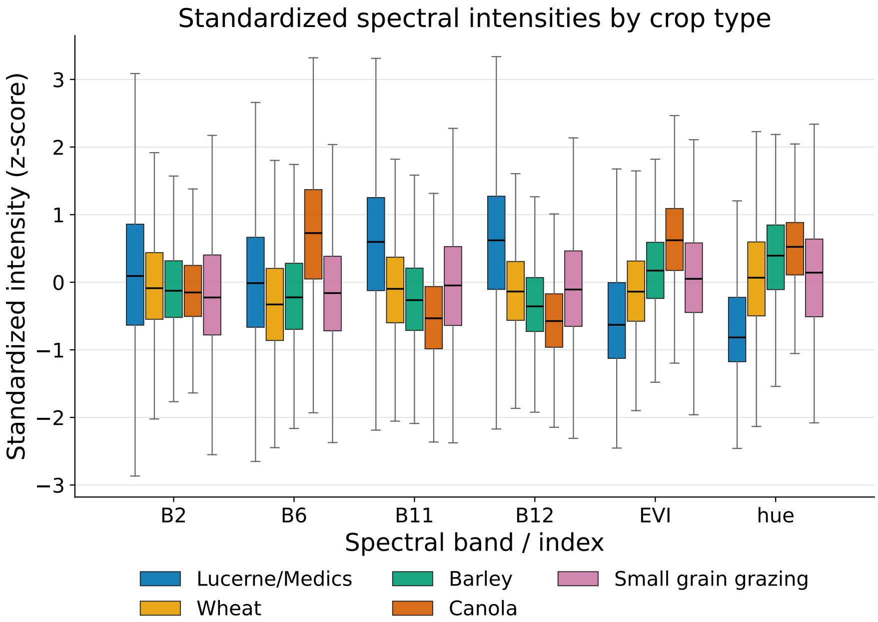

Fig. 3: Boxplots of spectral band intensities for each crop type, showing the distribution and overlap of spectral responses.

of time-series features, enabling the construction of high-dimensional representations without manual feature design [37].

In contrast, deep learning follows a representation learning paradigm, where models learn hierarchical feature representations directly from raw input data. Convolutional Neural Networks (CNNs) capture local spatial and spectral patterns, while Recurrent Neural Networks (RNNs) [14] and Transformer-based architectures [33] model temporal dependencies and long-range interactions in multi-temporal data. These approaches reduce reliance on hand-crafted features and allow the model to learn complex crop growth dynamics directly from the data.

Beyond pixel-level modeling, patch-based approaches introduce an additional spatial dimension by processing contiguous regions of imagery rather than individual pixels. This enables models to capture local texture, spatial heterogeneity, and contextual relationships between neighboring pixels, which are particularly important in agricultural landscapes where crop patterns exhibit strong spatial coherence. Patch-based models therefore complement both classical and deep learning approaches by explicitly incorporating spatial context into the classification process.

This study systematically compares these paradigms to evaluate their relative strengths, computational trade-offs, and effectiveness in handling complex agricultural environments, particularly under conditions of spectral similarity and temporal variability.

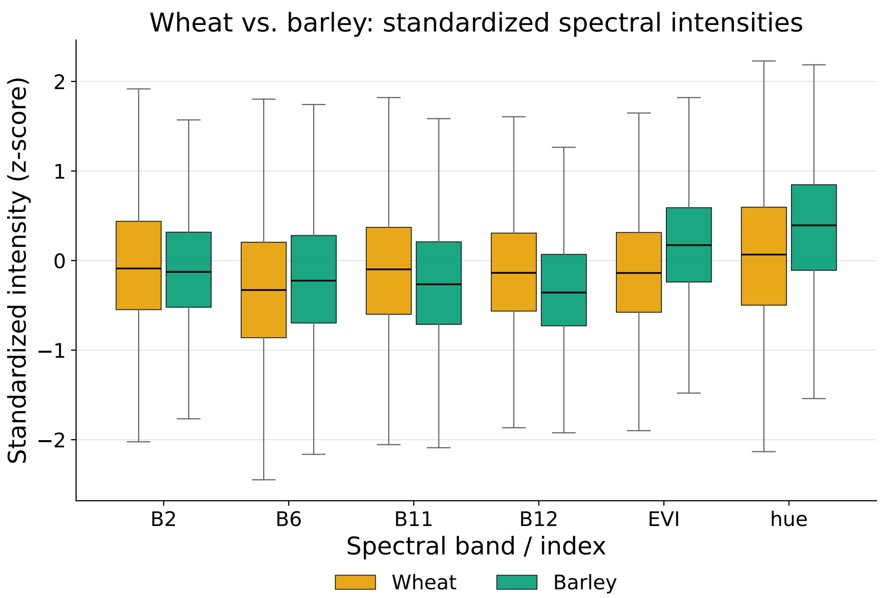

Fig. 4: Boxplots comparing spectral band intensities of wheat and barley, highlighting their high degree of spectral similarity and potential for classification confusion.

Model performance was assessed using a combination of standard classification metrics, including Cohen's Kappa, F1-score, and prediction entropy. These metrics were selected to evaluate not only overall accuracy but also class balance and prediction confidence across imbalanced crop types.

3.1 Classical Machine Learning

In the classical approach, we automated feature extraction using the `xr_fresh` toolkit to rapidly compute a comprehensive set of statistical and temporal features from each 10-month spectral time series [8, 37]. To address class imbalance, we applied SMOTE. Several classifiers such as Logistic Regression, Random Forest, LightGBM, and XGBoost were then evaluated.

Table 1 lists the time-series features extracted from each pixel's 10-month spectral profile [8].

This toolkit allowed us to extract all required time-series features efficiently, ensuring each pixel's temporal signature was well-represented for robust classification [37].

3.1.1 Pixel-Level Analysis

For pixel-level analysis, each pixel was treated as an independent sample. Feature vectors from each pixel's 10-month history were used to train classifiers. During inference, pixel predictions within a field polygon were aggregated via majority voting to determine the overall field label.

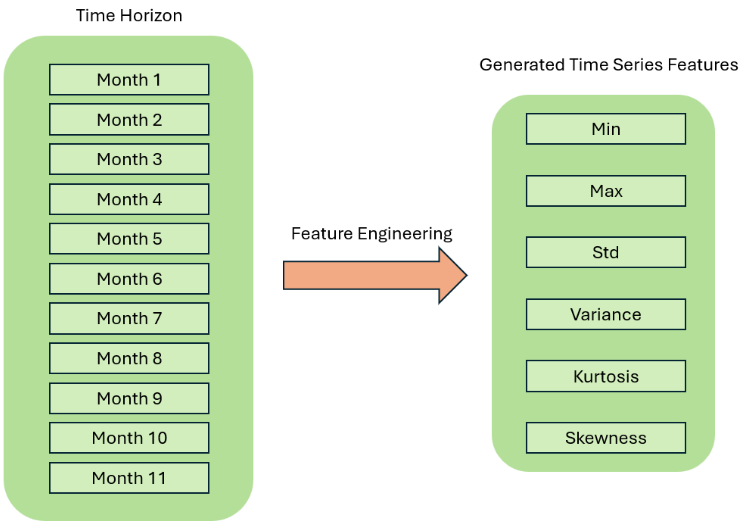

Fig. 5: Extraction of statistical and temporal features from the 10-month spectral time series.

Baseline Classical Models:

To benchmark crop classification, we used four classical models: Logistic Regression, Random Forest, LightGBM, and XGBoost—ranging from interpretable linear models to advanced tree-based learners.

Each pixel was represented by a feature vector $\mathbf{x}_i \in \mathbb{R}^{174}$, formed from time-series features across ten months and six spectral indices (B2, B6, B11, B12, EVI, Hue), capturing phenological signatures over time.

These four models are standard and we summarize only their configurations here; their formal definitions are collected in Appendix A. Logistic Regression [19] is a linear multinomial classifier providing interpretable per-class weights. Random Forest [20] is a bagged ensemble of decision trees whose axis-aligned splits capture non-linear interactions; we used 300 trees. LightGBM [21] and XGBoost [22, 23] are gradient-boosted tree ensembles with leaf-wise growth and explicit L_1/L_2 regularization; both were trained with GPU histogram methods. All four operate on the same per-field `xr_fresh` feature vector and, being tree- or margin-based with built-in feature selection or shrinkage, serve as the parsimonious baselines against which the deep models are compared.

Pixel-Level Ensemble with Hybrid Voting:

We ensembled RF, XGBoost, and LightGBM using soft voting. Per-pixel class probabilities were averaged:

$$\hat{\mathbf{P}}_i = \frac{1}{3}\left(\mathbf{P}_{\mathrm{RF}}(\mathbf{x}_i) + \mathbf{P}_{\mathrm{XGB}}(\mathbf{x}_i) + \mathbf{P}_{\mathrm{LGBM}}(\mathbf{x}_i)\right) \quad \hat{y}_i = \arg\max_k \hat{P}_i(k)$$

Table 1: Time-series features extracted from each pixel's spectral profile with `xr_fresh`.

Feature	Description
Absolute Energy	Sum of squared values, capturing overall signal magnitude.
Absolute Sum of Changes	Total absolute difference between consecutive measurements, reflecting temporal variability.
Autocorrelation (1, 2-month lag)	Correlation with its one- and two-month lagged versions, indicating series persistence.
Count Above Mean	Number of points above the series mean, quantifying high-value occurrences.
Day of Year of Max/Min	Julian day when the maximum and minimum occur, marking peak and low activity periods.
Kurtosis	Tail heaviness of the distribution, highlighting propensity for extremes.
Linear Trend	Slope of a least-squares fit, summarizing the overall upward or downward trend.
Longest Strike Above/Below Mean	Longest consecutive run above or below the mean, capturing sustained periods.
Max/Min	Absolute extreme values in the series.
Mean/Median	Measures of central tendency.
Mean Absolute Change/Mean Change	Average magnitude and signed change between consecutive points.
Mean Second Derivative	Mean of the series' second differences, detecting acceleration or deceleration.
Quantiles (0.05, 0.95)	Values at the 5th and 95th percentiles, summarizing distribution extremes.
Ratio Beyond $r\sigma$	Proportion of points more than r standard deviations ($r = 1, 2, 3$) from the mean.
Skewness	Asymmetry of the distribution around the mean.
Standard Deviation/Variance	Spread measures around the mean.
Sum of Values	Total sum of observations (mean $\times$ length).
Symmetry Looking	Similarity when the series is reversed.
Time Series Complexity	Complexity-Invariant Distance metric of sequence irregularity.

Field-Level Aggregation:

Let field f consist of pixels $i \in f$. Field-level soft voting is computed as:

$$\bar{\mathbf{P}}_f = \frac{1}{|f|} \sum_{i \in f} \hat{\mathbf{P}}_i \quad \hat{y}_f^{\text{soft}} = \arg\max_k \bar{P}_f(k)$$

We introduced a confidence margin:

$$\Delta_f = \bar{P}_f^{(1)} - \bar{P}_f^{(2)}$$

where $\bar{P}_f^{(1)}$ and $\bar{P}_f^{(2)}$ are the highest and second-highest class probabilities in $\bar{P}_f$, respectively.

If $\Delta_f < \theta$ (threshold = 0.10), we fall back to mode voting across pixel labels:

$$\hat{y}_f^{\text{mode}} = \text{mode}\,\{\hat{y}_i \mid i \in f\}$$

Final Hybrid Rule:

$$\hat{y}_f = \begin{cases} \hat{y}_f^{\text{soft}}, & \text{if } \Delta_f \geq \theta \\ \hat{y}_f^{\text{mode}}, & \text{otherwise} \end{cases}$$

This strategy balanced probabilistic confidence with categorical consistency, improving field-level classification stability.

3.1.2 Field-Level Analysis

In the field-level analysis, we aggregated all pixel features within each field polygon, as shown in Figure 6, by computing the mean across pixels to form a single representative feature vector per field. This step reduced noise and the influence of outlier pixels while preserving meaningful spatial patterns. Classifiers were then trained using these aggregated field-level vectors to directly learn from spatially coherent crop representations. This approach aligns with previous findings that object-based aggregation helps in minimizing spectral confusion and enhances classification accuracy, especially in homogeneous agricultural regions [11].

To address class imbalance at the field level, we combined SMOTETomek resampling with a stacked ensemble classifier. This approach ensured both balanced data and strong generalization performance.

SMOTETomek Resampling

SMOTETomek combines oversampling and undersampling techniques. SMOTE generates synthetic samples for underrepresented classes by interpolating between existing samples, while Tomek Links removes borderline examples to improve class separability. This helped mitigate the severe class imbalance visible in Figure 1, especially for crops like small grain grazing and canola.

Stacked Ensemble Classifier

We built a two-layer ensemble consisting of Random Forest and XGBoost as base learners, with Logistic Regression (using `passthrough=True`) as the meta-learner,

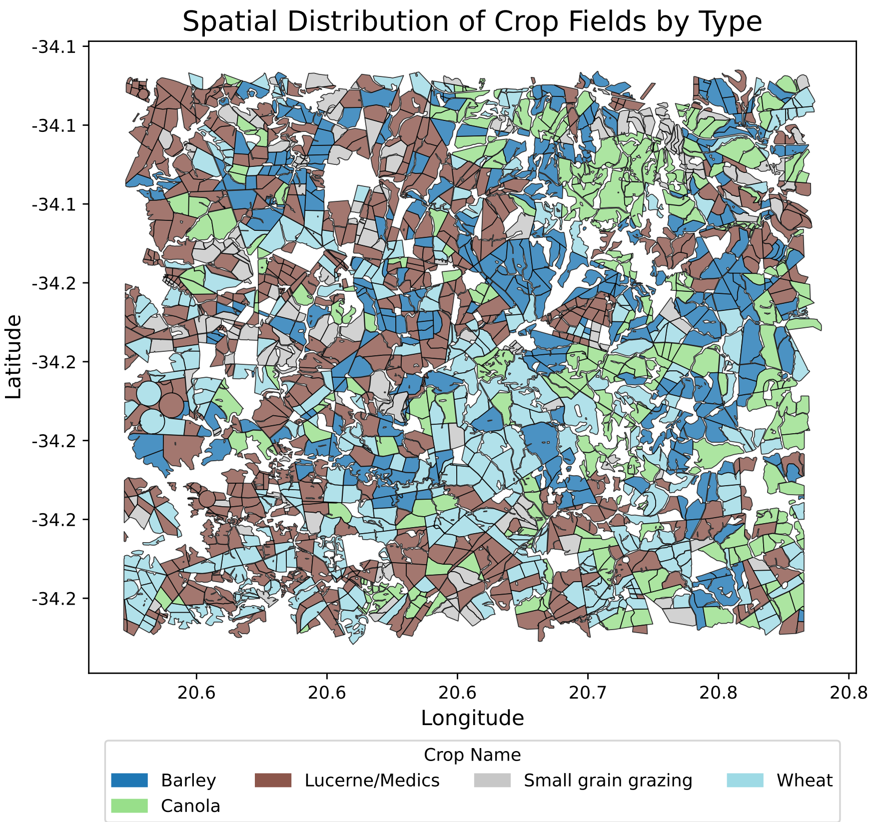

Fig. 6: Illustration of field-level aggregation: pixel values are grouped by field polygon and summarized into a single feature vector.

allowing it to use both base-model predictions and the original input features. The final prediction for pixel i is defined as:

$$\hat{y}_i^{\text{stacked}} = f_{\text{meta}}\left(f_{\text{RF}}(\mathbf{x}_i),\ f_{\text{XGB}}(\mathbf{x}_i),\ \mathbf{x}_i\right)$$

This architecture combines the non-linear modeling capacity of tree-based learners with a linear meta-learner that integrates both base predictions and original input features.

Field-Based Aggregation and XGBoost Optimization

We also implemented an optimized field-level pipeline using XGBoost. Field identifiers were used to ensure that no field contributed to more than one data split, with fields split into train/validation/test sets using `fid`-based stratification. Features were aggregated via the mean across pixels, and labels via the mode crop type per field.

Optuna was then used to tune XGBoost over 75 trials, searching across `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, `gamma`, `reg_alpha`, `reg_lambda`, and `min_child_weight`, with the objective of maximizing the weighted F1 score via 5-fold stratified cross-validation. The best configuration (Trial 32) is detailed in Table 2, which also achieved a validation accuracy of 76.61%.

Table 2: Top 5 trials from hyperparameter search with corresponding parameters and validation accuracies

Trial	**Acc.**	**Est.**	**Depth**	**LR**	**SubS**	**ByTree**	**Gamma**	**Regα**	**Regλ**	**MinCW**
32	**76.61%**	1141	13	0.0215	0.5749	0.6986	0.5589	0.00184	0.00259	5
12	76.46%	984	12	0.0145	0.6675	0.6946	0.0799	0.00503	0.00455	7
24	76.40%	1165	11	0.0162	0.7831	0.6403	0.00413	0.00481	0.00478	9
13	76.29%	995	12	0.0277	0.6345	0.6829	0.3999	0.00011	0.00567	6
18	76.29%	915	13	0.0121	0.6784	0.5958	0.1612	0.00491	0.1732	8

3.2 Deep Learning Architectures for Crop Classification

Deep learning models ingest raw spectral values and learn hierarchical representations that capture both spatial and temporal dependencies, offering an alternative to the manual feature engineering required by classical approaches [24, 25]. We explore architectures across two spatial granularities: pixel-level models that operate on per-pixel temporal sequences, and patch-level models that additionally exploit local spatial structure. To ensure unbiased evaluation across all architectures, all pixels belonging to a given field were assigned exclusively to either the training, validation, or test set, preventing spatial information leakage between splits.

3.2.1 Pixel-Level Architectures

At the pixel level, each sample is an individual pixel characterized by 60 temporal features spanning the 2017 growing season — six spectral bands and vegetation indices (B2, B6, B11, B12, EVI, Hue) observed across 10 monthly composites. Each pixel is linked to a unique field identifier and labeled according to the field crop type across five classes: wheat, barley, canola, small grain grazing, and lucerne/medics. Two consequences of this design should be kept in mind. First, pixels within a field are highly spatially autocorrelated and all inherit the same field label — including mixed pixels along field boundaries — so although the pixel-level models are trained on millions of samples, the number of statistically independent units is closer to the number of fields than the number of pixels, and the pixel models' apparent data volume overstates the effective training size. Second, to prevent leakage we perform all train/validation/test splitting at the field (`fid`) level, so no field's pixels straddle two partitions; this keeps evaluation honest at the field scale even though training treats pixels as independent. Pixel-level predictions are subsequently aggregated to the field level via majority voting:

$$\hat{y}_f = \mathrm{mode}(\{\hat{y}_i \mid i \in f\})$$

For TabNet models we applied z-score standardization; CNN-based models were trained on raw reflectance values, allowing convolutional layers to learn directly from temporal magnitude patterns.

CNN + BiLSTM Ensemble with Focal Loss.

To capture the intricate temporal dynamics of crop growth, we developed a hybrid architecture combining 1D Convolutional Neural Networks (CNNs) with Bidirectional Long Short-Term Memory (BiLSTM) networks (Figure 7).

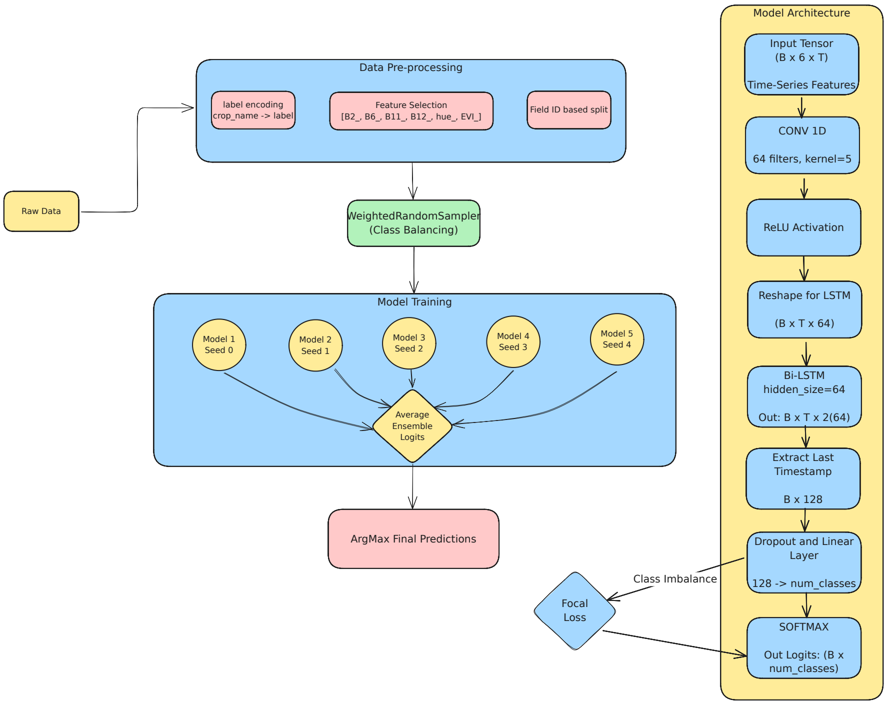

Fig. 7: Architecture of the CNN + BiLSTM model. The model combines 1D convolutional temporal feature extraction with sequential modeling through a BiLSTM layer.

The model processes an input tensor of shape $(B, 6, T)$. A 1D convolutional layer with 64 filters and kernel size 5 is applied along the temporal axis, followed by ReLU activation and tensor permutation to $(B, T, 64)$ for compatibility with the recurrent layer. A bidirectional LSTM with hidden size 64 then captures temporal dependencies in both forward and backward directions, yielding an output of shape $(B, T, 128)$. The representation at the final timestep passes through dropout and a fully connected layer, with class probabilities computed via softmax:

$$P(y_i = c \mid \mathbf{x}_i) = \frac{e^{z_{i,c}}}{\sum_{c'=1}^{C} e^{z_{i,c'}}}$$

To address class imbalance we used Focal Loss:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t)$$

where α_t is the class weighting factor and $\gamma = 2.0$ is the focusing parameter. A `WeightedRandomSampler` further ensured balanced class representation during training. To reduce variance, five identical models were trained with different random seeds and their logits averaged:

$$\text{Logits}_{\text{ensemble}} = \frac{1}{N} \sum_{k=1}^{N} \text{Logits}^{(k)}$$

Ensemble of TabNet Models.

TabNet treats satellite observations as structured tabular inputs — each pixel's repeated measurements organized as a flat feature vector — and learns through iterative attention-based feature selection rather than temporal ordering (Figure 8). At each decision step, an Attentive Transformer generates a sparse mask selecting the most relevant input dimensions, which are then transformed by a Feature Transformer. This enables the model to focus on different feature subsets at different depths without relying on explicit temporal structure [28, 27].

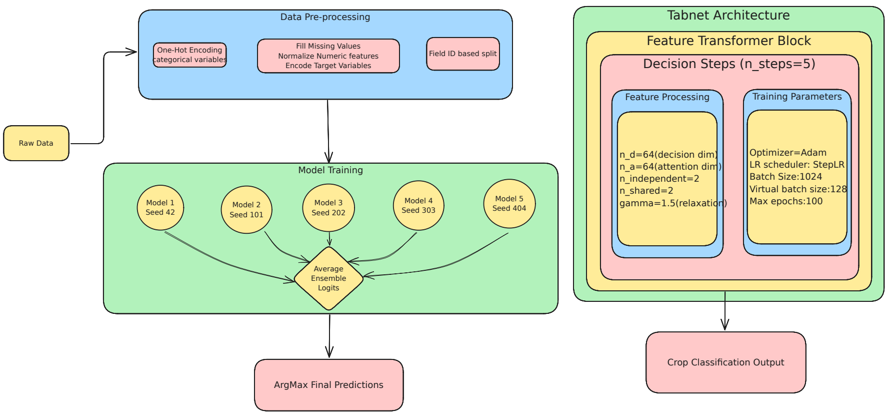

Fig. 8: Architecture of the TabNet ensemble. Each model applies iterative attention-based feature selection. Final predictions are obtained by averaging softmax outputs across five models, followed by field-level majority voting.

The network used $n_d = n_a = 64$, independent and shared block widths of 2, relaxation parameter $\gamma = 1.5$, and 5 decision steps. Training used the Adam optimizer with a StepLR scheduler, batch size 1024, virtual batch size 128, and early stopping monitored on a field ID-stratified validation set.

An ensemble of five TabNet models, each initialized with a distinct random seed (42, 101, 202, 303, 404), averaged softmax outputs to yield final class probabilities:

$$\hat{\mathbf{p}}_{\text{ensemble}} = \frac{1}{5}\sum_{i=1}^{5} \text{Softmax}\left(\mathbf{z}^{(i)}\right), \qquad \hat{y} = \arg\max_{c} \hat{p}_{\text{ensemble},c}$$

A practical advantage of TabNet is interpretability: the sparse decision masks can be inspected post hoc to identify which spectral bands or vegetation indices most influenced each prediction [27], an instance of the post-hoc explainability increasingly emphasized in remote sensing [39].

1D CNN with Optuna Hyperparameter Tuning.

As a lightweight baseline we implemented a 1D CNN whose architecture was optimized via Optuna [31] (Figure 9). The input is reshaped to $(B, 6, T)$, representing six spectral channels across monthly intervals, and passed through a single convolutional layer followed by ReLU, adaptive average pooling, dropout, and a fully connected classification head. The hyperparameter optimization objective was:

$$\theta^* = \arg\max_{\theta \in \mathcal{H}} \mathcal{A}_{\text{val}}(f_\theta)$$

where $\theta = \{\text{lr, dropout, filters, kernel size}\}$, solved using the Tree-structured Parzen Estimator (TPE). Each trial was evaluated over 25 epochs with a class-weighted sampler. The best configuration (Table 3) used a kernel size of 7, 64 filters, dropout ≈ 0.49, and learning rate 0.00116, yielding a validation accuracy of 75.5%.

Table 3: Top Optuna trials for the 1D CNN ranked by validation accuracy.

Trial	Val. Acc.	LR	Dropout	Filters	Kernel
14	**75.53%**	0.00116	0.5	64	7
13	74.06%	0.00102	0.5	128	7
19	74.61%	0.00118	0.4	64	7
7	73.20%	0.00013	0.3	128	7
11	73.66%	0.00006	0.4	128	7

Lightweight Temporal Attention Encoder (L-TAE).

The L-TAE applies self-attention directly to the temporal sequence of per-pixel reflectance vectors, learning which observations in the season are most informative

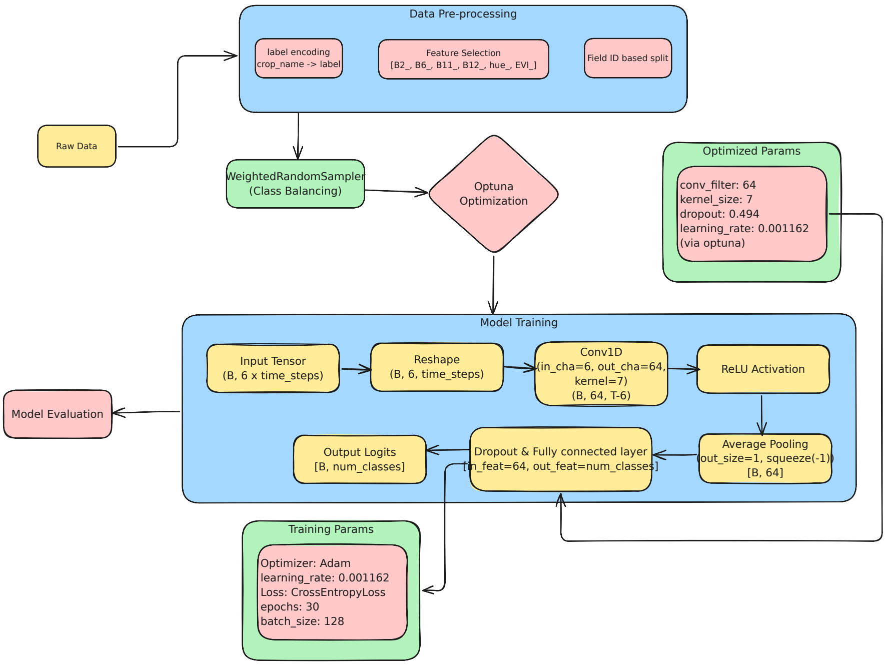

Fig. 9: Architecture of the 1D CNN model optimized via Optuna: a single convolutional layer followed by ReLU activation, adaptive pooling, and a fully connected classification head.

[29]. Given the embedded sequence $\{\mathbf{z}_t\}_{t=1}^{T}$, a set of learned master query vectors attend over time; for a single head the output is a temporally weighted sum

$$\mathbf{o} = \sum_{t=1}^{T} \alpha_t \, \mathbf{v}_t, \qquad \alpha_t = \frac{\exp\big(\mathbf{q}^\top \mathbf{k}_t / \sqrt{d}\big)}{\sum_{t'=1}^{T} \exp\big(\mathbf{q}^\top \mathbf{k}_{t'} / \sqrt{d}\big)},$$

where $\mathbf{k}_t, \mathbf{v}_t$ are key/value projections of $\mathbf{z}_t$, $\mathbf{q}$ is the learned query, and d is the head dimension. Channel grouping keeps the encoder lightweight. We evaluated L-TAE both at the pixel level and at the field level (averaging pixel sequences within a field before encoding), and trained a five-seed ensemble.

Temporal Convolutional Network (TempCNN).

TempCNN classifies a pixel from stacked one-dimensional convolutions applied along the temporal axis, each capturing local phenological motifs of increasing receptive field [30]. A convolutional layer computes

$$h_t^{(l)} = \mathrm{ReLU}\Big(\mathrm{BN}\left(\textstyle\sum_{\tau=-k}^{k} W_\tau^{(l)} \, h_{t+\tau}^{(l-1)} + b^{(l)}\right)\Big),$$

with batch normalization (BN) and dropout for regularization, followed by a fully connected softmax head. As with L-TAE, we evaluated pixel- and field-level variants and ensembled across five seeds.

3.2.2 Patch-Level Architectures

To exploit local spatial structure, we transitioned to a patch-level analysis in which each field is tiled into 100 m square patches ($\approx 10\times 10$ pixels at the 10 m resolution), each retaining the full 10-month spectral history for every pixel (Figure 10). Fields smaller than one hectare are kept whole as a single patch, and grid patches not fully contained within a field are discarded to avoid mixed-boundary pixels. Because the median field (7–11 ha) is several times larger than a patch, most fields yield several patches, providing implicit data augmentation. Before entering the convolutional models, each patch is resampled to a fixed 128×128 grid; we note that this upsamples the native $\sim 10\times 10$-pixel patches by roughly an order of magnitude, so the enlarged input reflects interpolation rather than additional spatial detail.

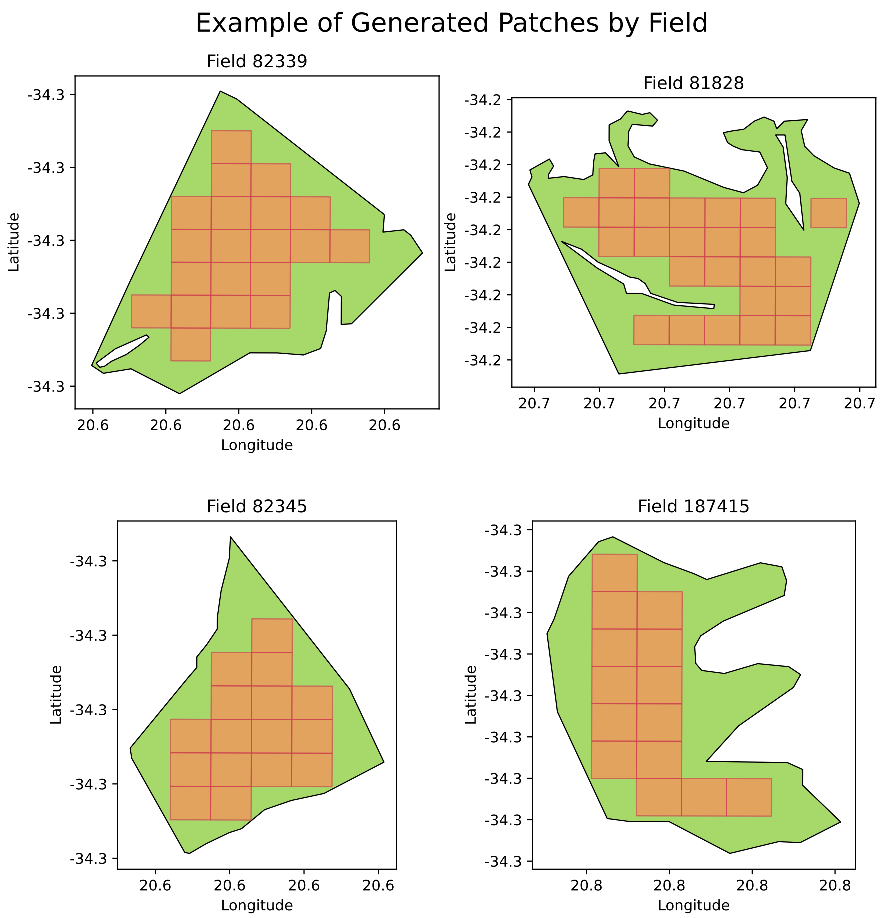

Fig. 10: Illustration of patch generation per field.

We investigated three distinct patch-based architectures.

Multi-Channel 2D CNN.

Each spectral band and monthly observation is treated as a separate input channel, giving a patch tensor $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ ($H = W = 128$ after resampling, $C = 60 = 6$ features $\times$ 10 months). Two Conv2D blocks with K filters each compute:

$$Y_k[i,j] = \mathrm{ReLU}\left(\sum_{c=1}^{C}\sum_{m=-1}^{1}\sum_{n=-1}^{1} W_{k,c,m,n}\cdot X[i+m,\, j+n,\, c] + b_k\right)$$

The feature map is flattened to $\mathbf{z} \in \mathbb{R}^{128}$ and passed to a fully connected classification head:

$$\hat{\mathbf{p}} = \mathrm{softmax}(W_{\mathrm{fc}}\,\mathbf{z} + b_{\mathrm{fc}})$$

Ensemble of Class-Specific 3D CNNs.

Five class-specific 3D CNNs are trained, each producing a scalar score $p_i = f_i(\mathbf{X})$ for its target class. A logistic regression meta-learner combines these scores into a final prediction (Figure 11):

$$\hat{\mathbf{p}} = \mathrm{softmax}\big(W_{\mathrm{meta}}[p_1, \ldots, p_5]^\top + b_{\mathrm{meta}}\big)$$

By modelling volumetric relationships jointly across spectral bands, spatial dimensions, and time, 3D convolutions capture crop-specific texture and phenological trajectories that 2D methods treat independently.

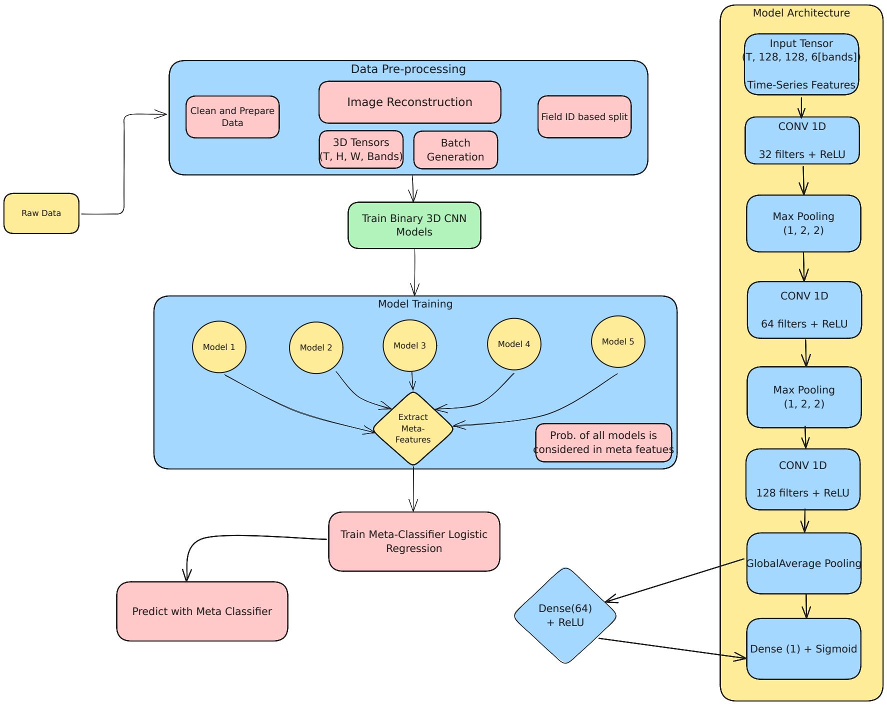

Fig. 11: 3D CNN ensemble architecture for patch-level crop classification.

Transformer-Based Model.

Each patch is represented as a sequence of multi-temporal image tensors of shape (T, H, W, B), where T is the number of temporal observations, $H \times W$ is the spatial resolution (128×128), and B is the number of spectral bands and indices.

For each time step, a lightweight convolutional encoder is applied using a 1×1 convolution followed by global average pooling to produce a compact feature vector $\mathbf{z}_t \in \mathbb{R}^d$, where $d = 32$ in our implementation. This results in a temporal sequence of patch-level embeddings $\{\mathbf{z}_t\}_{t=1}^{T}$.

To incorporate temporal order, learned positional embeddings are added:

$$\tilde{\mathbf{z}}_t = \mathbf{z}_t + \mathbf{e}_t$$

The resulting sequence is processed using stacked Transformer encoder blocks, each consisting of multi-head self-attention, residual connections, dropout, and a position-wise feedforward network:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

where $Q = \mathbf{Z}W_Q$, $K = \mathbf{Z}W_K$, and $V = \mathbf{Z}W_V$.

Finally, the encoded sequence is aggregated via global average pooling across time and passed through a fully connected layer with softmax activation to produce class probabilities:

$$\hat{\mathbf{p}} = \text{softmax}(W \cdot \mathbf{h} + b)$$

This architecture emphasizes temporal dependency modeling across monthly patch representations, while spatial information is incorporated indirectly through the convolutional preprocessing stage.

3.3 Performance Metrics

Because the class distribution is severely imbalanced (Figure 1), plain accuracy is misleading — a classifier can score highly simply by predicting the majority class. We therefore adopt macro-averaged F1 (F1_m), the unweighted mean of the per-class F1 scores, as our primary metric: it weights every crop equally and so directly penalizes failure on the minority cereals, which is exactly where models differ. We report three further metrics alongside it: Cohen's Kappa, chance-corrected agreement between predicted and true labels; weighted F1, the support-weighted mean of per-class F1; and log-loss (cross-entropy), the mean negative log-probability assigned to the true class (lower is better, capturing prediction confidence). Higher Kappa and F1 indicate better, more balanced classification. Formal definitions of all four metrics are given in Appendix B.

4 Results

We evaluated every model under two protocols. *In-region* performance uses a field-wise (`fid`-stratified) train/validation/test split drawn entirely from the training tiles — the conventional way crop classifiers are benchmarked. *Out-of-sample* (OOS) performance applies the trained model to the spatially disjoint holdout tile (`34S_20E_259N`; 2,417 fields). All predictions are aggregated to the field level. Because the classes are heavily imbalanced (Section 3.3), we report macro-averaged F1 (F1_m) as the primary metric, with Cohen's Kappa, weighted F1, and log-loss (entropy) alongside.

4.1 In-region validation overstates performance and misranks models

Table 4 reports both protocols for all individual models, ranked by out-of-sample macro-F1, with the generalization gap $\Delta = \mathrm{F1}_m^{\mathrm{OOS}} - \mathrm{F1}_m^{\text{in-region}}$ shown alongside; Figure 12 visualizes the same data ordered by Δ. Two facts stand out. First, *every* model loses accuracy out of sample, but the loss is highly uneven: it ranges from -0.05 (logistic regression) to -0.26 (CNN-BiLSTM). Second, this unevenness reorders the models. Under in-region validation the dense temporal deep nets look strongest — CNN-BiLSTM, TempCNN, and L-TAE all reach $\mathrm{F1}_m \approx 0.72$–$0.78$, and conventional field-level evaluation (on which the original analysis of this dataset relied) singled out CNN-BiLSTM as the best model. On the holdout tile that conclusion collapses: CNN-BiLSTM falls to $\mathrm{F1}_m = 0.46$ (Kappa 0.32), the *largest* drop of any model and second-worst absolute score. The same fate befalls the 3D CNN patch ensemble (-0.22) and SMOTE-resampled stacking (-0.21).

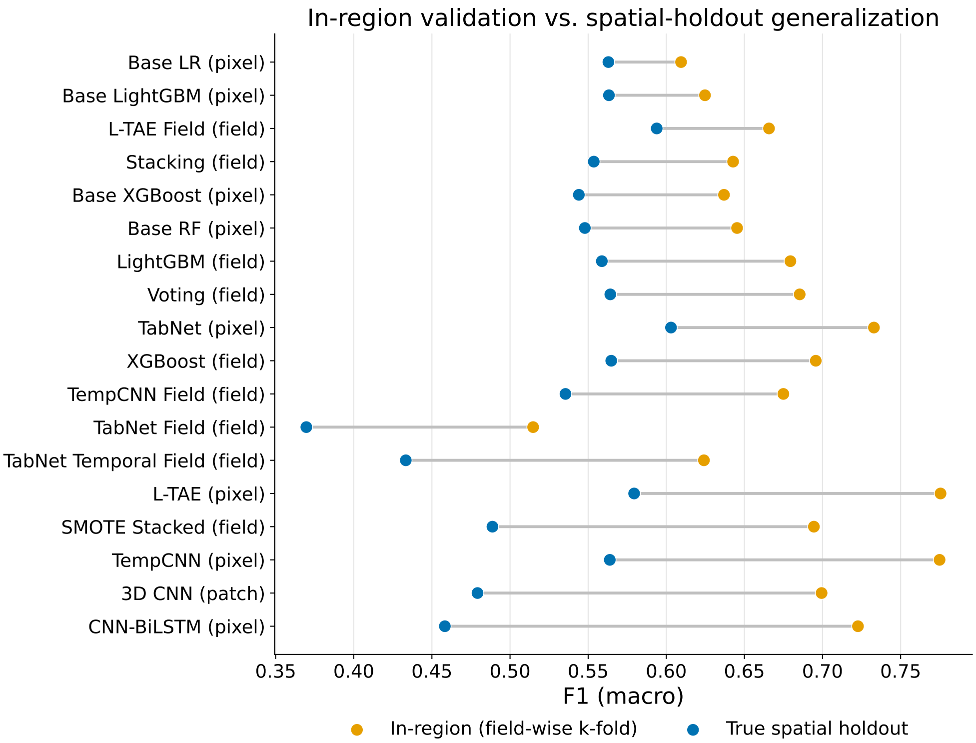

Fig. 12: In-region (field-wise cross-validation) versus spatial-holdout macro-F1 for each model, sorted by the gap. Dense temporal/patch deep nets (top) suffer the largest degradation; tree-based, linear, and field-aggregated models (bottom) transfer with the smallest gaps.

Table 4: In-region (field-wise CV) versus out-of-sample (spatial holdout) macro-F1 for every individual model, ranked by out-of-sample macro-F1 — the metric we argue model selection should use. In-region macro-F1 and the gap Δ are shown alongside (Figure 12 orders the same data by Δ); OOS Kappa, weighted F1 (wF1), and cross-entropy (Xent) complete the picture. The best transferer is a TabNet pixel ensemble, whereas the model conventional field-level evaluation singles out as best (CNN-BiLSTM) is among the worst; the highest in-region scores (L-TAE, TempCNN) do not yield the highest out-of-sample scores. The *Features* column marks whether a model is fed the automated `xr_fresh` time-series statistics (174 per field) or raw reflectance; all `xr_fresh` models are field-level classical learners.

			In-reg		**Out-of-sample (holdout)**			
Model	**Level**	**Features**	$\mathbf{F1}_m$	Δ	$\mathbf{F1}_m$	κ	**wF1**	**Xent**
TabNet	**pixel**	**raw**	**0.73**	−0.13	**0.60**	**0.54**	**0.70**	**4.42**
L-TAE	field	raw	0.67	−0.07	0.59	0.47	0.67	5.51
L-TAE	pixel	raw	0.78	−0.20	0.58	0.49	0.68	5.20
XGBoost	field	`xr_fresh`	0.70	−0.13	0.56	0.47	0.68	4.89
Voting	field	`xr_fresh`	0.69	−0.12	0.56	0.47	0.66	4.82
TempCNN	pixel	raw	0.77	−0.21	0.56	0.49	0.68	5.09
Base LightGBM	pixel	raw	0.62	−0.06	0.56	0.47	0.66	4.85
Logistic Regression	pixel	raw	0.61	−0.05	0.56	0.47	0.67	4.93
LightGBM	field	`xr_fresh`	0.68	−0.12	0.56	0.45	0.66	4.99
Stacking	field	`xr_fresh`	0.64	−0.09	0.55	0.47	0.67	5.01
Random Forest	pixel	raw	0.65	−0.10	0.55	0.45	0.65	5.07
Base XGBoost	pixel	raw	0.64	−0.09	0.54	0.47	0.66	5.01
TempCNN	field	raw	0.67	−0.14	0.54	0.43	0.64	6.07
SMOTE Stacked	field	`xr_fresh`	0.69	−0.21	0.49	0.43	0.63	5.37
3D CNN	patch	raw	0.70	−0.22	0.48	0.30	0.55	7.79
CNN-BiLSTM	pixel	raw	0.72	−0.26	0.46	0.32	0.56	7.73
TabNet Temporal	field	raw	0.62	−0.19	0.43	0.31	0.56	7.35
TabNet	field	`xr_fresh`	0.51	−0.15	0.37	0.25	0.53	6.13

The winners under spatial transfer are a different group entirely. The best single model is the TabNet pixel ensemble ($F1_m = 0.60$, Kappa 0.54), and the most *robust* models — those with the smallest gaps — are classical: logistic regression ($\Delta = -0.05$), pixel-level LightGBM (−0.06), and the tree-based field ensembles. Notably, the robust field ensembles in the upper-middle of the table (XGBoost, Voting, LightGBM, Stacking) are precisely the models fed the automated `xr_fresh` features (Table 4, Features column): the combination of a rich automated feature representation with a tree learner transfers well, whereas the deep nets that learn from raw sequences do not. A majority-vote ensemble over all models reaches $F1_m = 0.60$ (Kappa 0.52), essentially tying TabNet but not exceeding it, indicating that voting cannot rescue the fragile members. The practical implication is immediate: a practitioner who selected a model by in-region cross-validation would have deployed the single worst transferer.

4.2 Pixel versus field level: aggregation as variance reduction

Several architectures were run at both the pixel level and the field level (averaging the per-pixel time series within each field before classification). Table 5 and Figure 13 show that field-level aggregation acts as a variance-reduction regularizer for the *dense* temporal networks: L-TAE improves from $F1_m = 0.58$ (pixel) to 0.59 (field) while its gap shrinks from -0.20 to -0.07, and TempCNN's gap narrows from -0.21 to -0.14. Averaging pixels within a field cancels region-specific pixel noise before the model sees it, lowering the in-region ceiling but improving transfer. TabNet is the informative exception: aggregating its raw inputs to the field level *hurts* ($F1_m$ falls from 0.60 to 0.43), because its robustness already derives from sparse feature selection and field-level training starves it of the per-pixel sample volume it exploits (millions of pixels versus a few thousand fields). The tree ensembles transfer well at either level. In short, dense models need aggregation to generalize; sparse, tree-like models do not.

A second comparison is embedded in the same table. For the tree learners, the field-level model is fed `xr_fresh` statistics while the pixel-level model is fed raw reflectance, so their pixel→field change is exactly the effect of swapping raw inputs for the automated `xr_fresh` representation. That change is essentially neutral out of sample — XGBoost moves from $F1_m = 0.54$ to 0.56 and LightGBM stays at 0.56 — even though `xr_fresh` raises the in-region score (e.g. XGBoost 0.64 → 0.70) and therefore widens the gap. The 174 engineered features thus let a tree fit the training region more tightly without transferring any better; the value of `xr_fresh` here is operational — a single compact vector per field, trained in minutes without handling millions of pixels — rather than a gain in generalization accuracy. TabNet, uniquely, was run with *both* field representations and lets us separate the two effects: moving its raw inputs from pixel to field level costs $F1_m$ 0.60 → 0.43 (the aggregation/sample-volume penalty), and substituting `xr_fresh` for the raw field-averaged input costs a further 0.43 → 0.37. So for TabNet neither aggregation nor `xr_fresh` helps — consistent with the conclusion that `xr_fresh` buys operational convenience rather than transfer accuracy.

4.3 Data efficiency under spatial transfer

To probe how much labeled data the transferable models actually require, we retrained a representative set on random subsets of 75%, 50%, and 25% of the training fields and re-evaluated on the holdout tile (Figure 14). The most robust models are also the most data-efficient: TabNet retains essentially full holdout accuracy down to a quarter of the fields ($F1_m = 0.60$, 0.62, 0.62, 0.59 at 100/75/50/25%), and logistic regression is similarly flat (0.56–0.60). The dense pixel L-TAE degrades more steeply (0.58 to 0.57), and the explicit feature-selection network LassoNet is poor at every budget ($F1_m \approx 0.46$). Holdout accuracy is therefore limited far more by spatial transfer than by training-set size for the robust models — halving the labels costs them little.

4.4 Feature dimensionality and redundancy

A natural concern with the 174-dimensional `xr_fresh` representation is feature redundancy. We tested three forms of explicit dimensionality control. *LassoNet*, which learns a hard feature-selection mask, generalized worst of all pixel models ($F1_m \approx 0.46$

Table 5: Out-of-sample macro-F1 (and gap Δ) at pixel versus field level for architectures evaluated at both granularities. The pixel input is raw reflectance in every case; the *Field input* column gives the field-level representation — either the automated `xr_fresh` statistics or the field-averaged raw time series (raw-avg). For the tree learners (XGBoost, LightGBM) the pixel→field change is therefore precisely a raw→`xr_fresh` change. Field aggregation helps the dense temporal nets and is roughly neutral for the tree models, but degrades TabNet.

	Pixel		**Field**		
Architecture	**OOS F1**$_m$	**Δ**	**OOS F1**$_m$	**Δ**	**Field input**
L-TAE	0.58	−0.20	0.59	−0.07	raw-avg
TempCNN	0.56	−0.21	0.54	−0.14	raw-avg
TabNet	0.60	−0.13	0.43	−0.19	raw-avg
XGBoost	0.54	−0.09	0.56	−0.13	`xr_fresh`
LightGBM	0.56	−0.06	0.56	−0.12	`xr_fresh`

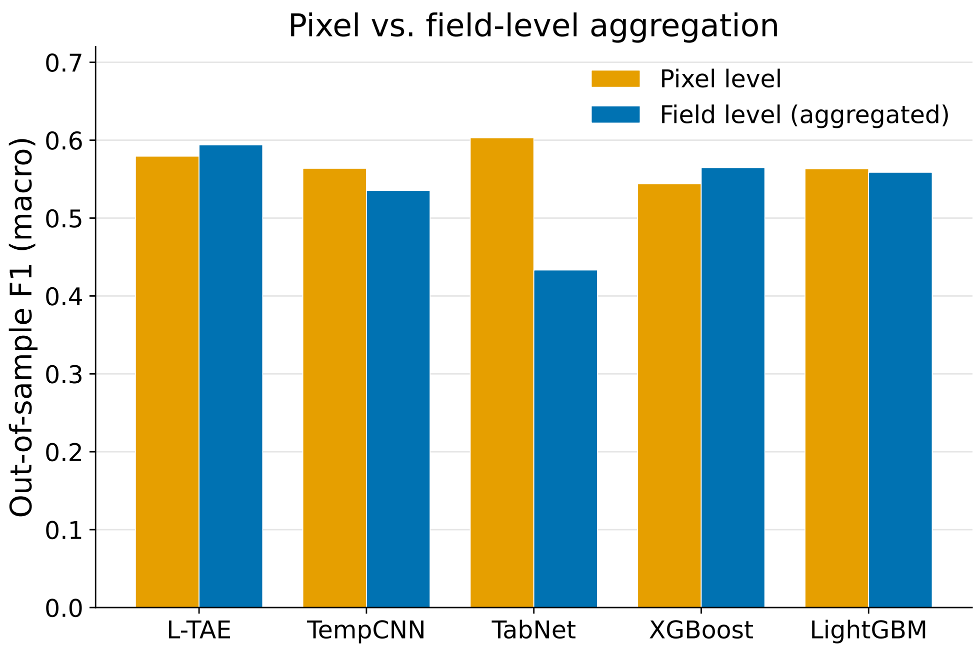

Fig. 13: Out-of-sample macro-F1 at pixel versus field level. Field-level aggregation recovers transfer for dense temporal models (L-TAE) but collapses the sparsity-reliant TabNet.

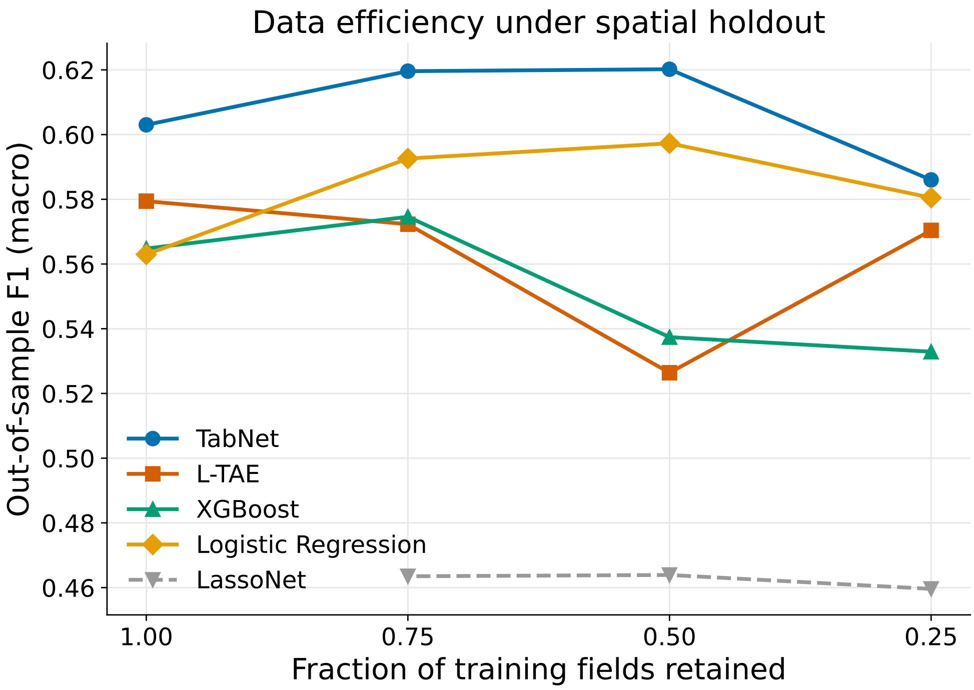

Fig. 14: Out-of-sample macro-F1 as the number of training fields is reduced to 75%, 50%, and 25%. Robust models (TabNet, logistic regression) are nearly flat; LassoNet is weak throughout.

at every training budget; Figure 14), indicating that aggressive pruning discards transfer-relevant signal. A *sparse-attention* L-TAE variant (TabNet-style channel gating) modestly improved transfer over the dense L-TAE at matched data (0.61 vs. 0.57 at 75% of fields), consistent with *soft* sparsity helping. L_2 *regularization* of the tree models changed OOS $\mathrm{F1}_m$ negligibly (within ± 0.01). Together these show that the high feature dimensionality is not the binding constraint — soft, learned sparsity helps a little, hard selection hurts, and the dominant limitation remains spatial transfer.

4.5 Per-class performance and persistent confusions

Figure 15 breaks out per-class holdout F1 for the best transferer (TabNet), a robust tree model (XGBoost field), a robust linear baseline (logistic regression), and the largest faller (CNN-BiLSTM). The gap is highly class-dependent. All models classify the majority crop lucerne/medics and the spectrally distinct canola well (TabNet F1 0.88 and 0.81 respectively), but accuracy on the minority cereals is poor and is where the transfer loss concentrates: TabNet reaches only 0.55 on barley, 0.48 on wheat, and 0.30 on small grain grazing. CNN-BiLSTM underperforms across *all* classes, not just the rare ones, confirming a generalization rather than an imbalance failure. The wheat–barley and small-grain-grazing confusions are consistent with their spectral overlap (Figure 4) and persist across every model and paradigm.

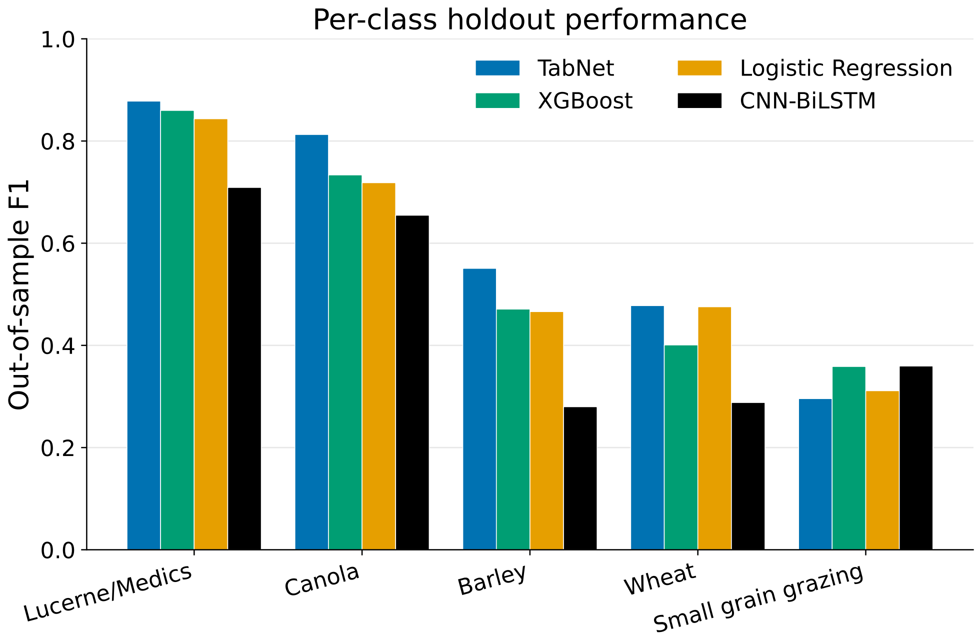

Fig. 15: Per-class out-of-sample F1 for four representative models. Lucerne/medics and canola are classified well by all; barley, wheat, and small grain grazing concentrate the errors. CNN-BiLSTM degrades across every class.

5 Discussion

Our results demonstrate that, in this setting, the conclusion one draws about which model is "best" depends almost entirely on whether evaluation is in-region or spatially out-of-sample — and that a model's transfer behaviour is predictable from its inductive bias.

Why tree-like inductive bias transfers.

The single clearest pattern in our results is that transfer tracks *inductive bias*, not model family or capacity per se. Figure 16 groups every model by inductive bias and plots its generalization gap. The ordering is stark: linear models lose the least (mean $\Delta = -0.05$), tree-based models little (mean -0.11), while dense temporal/patch deep nets lose the most (mean -0.22) and SMOTE-augmented stacking is similarly fragile (-0.21). Random Forest and gradient-boosted trees have been the remote-sensing workhorse for two decades precisely because their sparse, axis-aligned splits select a few informative features at each node and ignore the rest, which resists overfitting to the idiosyncratic, region-specific structure of any one scene [12, 24]. Our results show that this same property is what governs *spatial* transfer. Crucially, the best out-of-sample model, TabNet, is a neural network built explicitly to emulate this behaviour: its Attentive Transformer emits sparse (sparsemax) feature masks that perform instance-wise, axis-aligned feature selection at each decision step, mirroring tree splits while remaining differentiable [28]. TabNet thus clusters with the trees, not with the dense sequence models, and inherits their transferability. By contrast, CNN-BiLSTM, TempCNN, and L-TAE build dense distributed representations over the full temporal sequence with no built-in feature sparsity; they fit the training region's phenology and atmosphere tightly and transfer poorly.

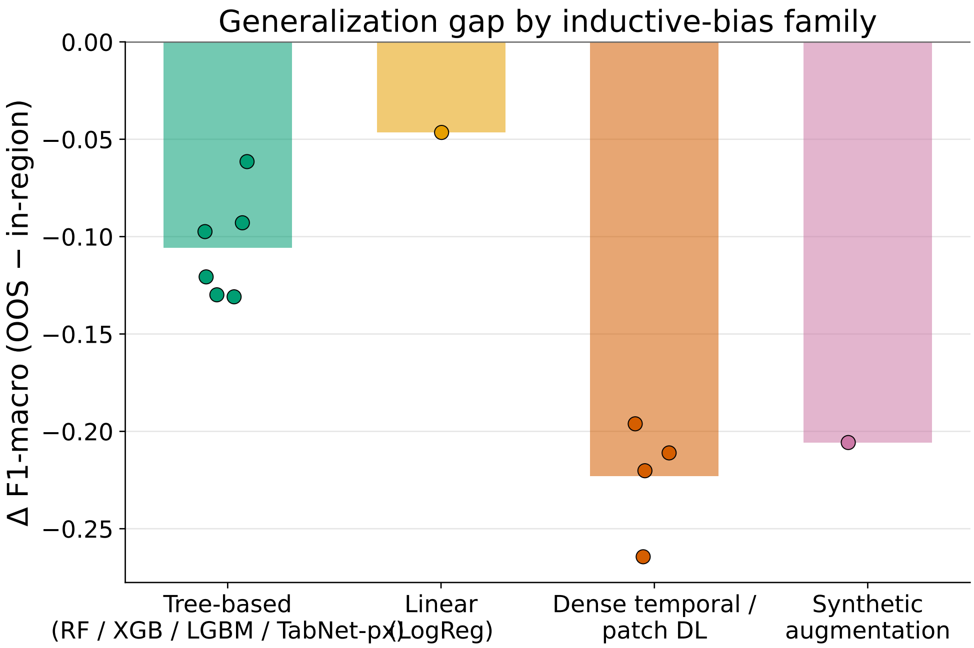

Fig. 16: Generalization gap (Δ macro-F1, out-of-sample minus in-region) grouped by inductive-bias family. Sparse/tree-like and linear models transfer (small |Δ|); dense temporal/patch deep nets and synthetic augmentation overfit the training region. Bars show family means; points are individual models.

Two robustness levers.

We are careful not to overclaim a single tidy axis, because two caveats sharpen rather than weaken the story. First, it is *pixel-level* TabNet that wins: the field-aggregated TabNet variants are among the worst transferers (Table 4), because aggregation removes the per-pixel sample volume that the sparse model exploits. Second, L-TAE

— an attention model — transfers well *when* applied at the field level ($\Delta = -0.07$). These observations point to two distinct, partly substitutable routes to robustness. The first is a sparse, tree-like inductive bias baked into the model (trees, TabNet), which confers transfer intrinsically. The second is variance reduction imposed from outside via field-level aggregation (Section 4.2), which averages away region-specific pixel noise and rescues otherwise-fragile dense models. A model that already possesses the first lever gains nothing — and may lose sample volume — from the second, which is exactly why TabNet prefers pixels and L-TAE prefers fields. The fragile cluster (CNN-BiLSTM, 3D CNN, SMOTE stacking) combines high capacity, dense temporal representation, and (for SMOTE) synthetic interpolation, with neither lever engaged.

Automated feature extraction and computational cost.

The classical pipelines paired with `xr_fresh` were not only robust but cheap: the field-level Voting and Stacking ensembles transfer as well as the best deep models (OOS $F1_m$ 0.55–0.56) while training in under two minutes, versus hours for the temporal deep nets (Table 4). To our knowledge this is the first evaluation of `xr_fresh` in a crop-classification context, and the result is encouraging: automated time-series feature extraction yields a representation that, fed to a tree ensemble, both matches deep models on transfer and inherits the trees' robustness — a favourable combination for operational, resource-constrained deployment.

Patch-level spatial modeling.

Patch-level 3D CNN models did not exceed pixel-level models in-region and transferred poorly ($F1_m = 0.48$, $\Delta = -0.22$). This is partly structural. At Sentinel-2's 10 m resolution the native 100 m patches are only $\sim 10 \times 10$ pixels, so the local spatial context they can encode is limited, and resampling them to the network's 128×128 input adds interpolated rather than genuine detail. Narrow or small fields yield few fully-contained interior patches, further limiting the spatial signal available. Combined with a dense convolutional representation that offers no transfer-protective sparsity, these factors leave the patch models with little to exploit and the weakest out-of-sample performance of any paradigm.

Persistent difficulty with small grain grazing and barley.

Small grain grazing, barley, and wheat are the hardest classes for every model and paradigm (Figure 15), and they concentrate the out-of-sample loss. Small grain grazing has both the fewest distinctive observations and a spectral-temporal signature that closely resembles the other cereals, particularly early in the season; wheat–barley confusion is similarly persistent and consistent with their spectral overlap (Figure 4). That even the robust models struggle here — and that CNN-BiLSTM degrades across *all* classes, not only the rare ones — indicates the limitation is partly intrinsic class similarity and partly spatial transfer, not class imbalance alone. Targeted collection for these classes, and richer temporal sampling, are the most promising remedies.

Limitations.
Several limitations constrain these results. First, the study uses a single growing season (2017) and a single holdout tile that is spatially *adjacent* to the training tiles and lies in the same agroecological zone and growing calendar. This is therefore a relatively local spatial shift, and the gaps we report most likely *understate* the degradation that fully cross-regional or cross-year transfer would produce — our numbers are best read as a conservative lower bound. That even this mild shift reorders the models is precisely the cautionary point, but multi-region and multi-year holdouts are needed to confirm that the inductive-bias ordering is stable across larger distribution shifts; we accordingly frame our finding as a method-level conclusion about transfer rather than a claim about absolute achievable accuracy on this small benchmark. Second, the labels derive from a public Radiant Earth MLHub dataset rather than our own field campaign; this aids reproducibility but means we inherit its digitizing and survey conventions. Third, we use Sentinel-2 optical data only. The exclusion of SAR is a deliberate scoping choice — to isolate the optical time-series signal and maximize applicability where only optical data exist — but it limits the achievable ceiling, particularly for optically similar cereals; SAR fusion is a clear next step (below).

6 Conclusion

This study evaluated nineteen classical, deep learning, and patch-based crop classifiers in the Western Cape of South Africa under two protocols: conventional field-wise cross-validation within the training region, and a spatially disjoint holdout tile. The central finding is that these protocols disagree sharply about which model is best. Under in-region validation the dense temporal deep nets lead, and conventional field-level evaluation singles out a CNN-BiLSTM ensemble as best (macro-F1 0.72); under spatial transfer CNN-BiLSTM suffers the largest degradation of any model (macro-F1 0.46) and is among the worst, while a TabNet ensemble becomes the best single model (macro-F1 0.60, Kappa 0.54) and simple classical baselines — gradient-boosted trees and logistic regression — transfer with the smallest gaps.

We argue this reordering is mechanistic rather than incidental: the sparse, axis-aligned feature selection that made tree ensembles the remote-sensing default is precisely the inductive bias that confers spatial transfer, and TabNet inherits it through attentive sparsemax feature masks while dense temporal deep nets do not. Field-level aggregation provides a second, substitutable robustness lever — variance reduction that rescues dense models but is redundant for sparse ones. The most transferable models are also the most data-efficient, retaining near-full holdout accuracy on a quarter of the training fields.

For the remote-sensing community the practical message is concrete: report spatial-holdout performance, not just within-region cross-validation, because the latter is optimistic and can invert model rankings; prefer parsimonious, feature-selecting models (tree ensembles, TabNet) when the goal is mapping unseen terrain; and use field-level aggregation to harden dense models when they are required. Across all models, small-grain-grazing and wheat–barley confusion remain the dominant residual errors, calling for richer temporal sampling and targeted collection for these classes.

The most productive future directions are SAR–optical fusion to raise the achievable ceiling for spectrally similar cereals, multi-region and multi-year holdouts to test the stability of the inductive-bias ordering, and domain-adaptation methods that explicitly optimize for transfer. Higher-cadence optical platforms such as PlanetScope could further mitigate the cloud-cover and temporal-resolution limits encountered here.

Conflict of Interest

The authors declare that they have no competing interests.

Data Availability

This study relied on multiple sources of data. Ground-truth crop type labels were obtained from Radiant Earth's MLHub platform [2], while multi-temporal Sentinel-2 imagery was extracted using the Google Earth Engine API. Since the raw data was processed and combined in various ways specific to this study, the resulting dataset used for model training and evaluation is not hosted publicly but can be shared by the authors upon reasonable request for academic or research purposes.

Code Availability

All source code used for data processing, modeling, and evaluation is publicly available [3].

A Baseline classical model definitions

For completeness we reproduce the standard formulations of the four baseline classifiers described in Section 3.1. For a feature vector $\mathbf{x}_i$ and class $c \in \{1, \ldots, C\}$:
Logistic Regression [19] (multinomial / softmax):

$$P(y_i = c \mid \mathbf{x}_i) = \frac{\exp(\mathbf{w}_c^\top \mathbf{x}_i + b_c)}{\sum_{c'=1}^{C} \exp(\mathbf{w}_{c'}^\top \mathbf{x}_i + b_{c'})},$$

with per-class weight vectors $\mathbf{w}_c$ and biases b_c.
Random Forest [20]: majority vote over K decision trees,

$$\hat{y}_i = \mathrm{mode}\,\{T_1(\mathbf{x}_i), \ldots, T_K(\mathbf{x}_i)\}\,.$$

LightGBM [21]: additive gradient boosting with learning rate η and tree f_t,

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta \cdot f_t(\mathbf{x}_i).$$

XGBoost [22, 23]: regularized boosting objective,

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \qquad \Omega(f_k) = \gamma T_k + \tfrac{1}{2}\lambda \sum_j w_j^2,$$

where l is the per-sample loss, T_k the number of leaves in tree f_k, w_j the leaf weights, and γ, λ regularization coefficients.

B Performance metric definitions

Let N be the number of samples, C the number of classes, and n_c the support of class c. Cohen's Kappa is

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad p_o = \frac{\#\text{correct}}{N}, \quad p_e = \sum_{c=1}^{C} \frac{n_{c,\text{true}}}{N} \cdot \frac{n_{c,\text{pred}}}{N}.$$

Macro-averaged F1 (our primary metric) and weighted F1 are

$$\text{F1}_m = \frac{1}{C} \sum_{c=1}^{C} \text{F1}_c, \qquad \text{F1}_{\text{weighted}} = \sum_{c=1}^{C} \frac{n_c}{N} \, \text{F1}_c, \qquad \text{F1}_c = \frac{2 \, \text{Prec}_c \, \text{Rec}_c}{\text{Prec}_c + \text{Rec}_c}.$$

Log-loss (cross-entropy), with one-hot labels $y_{i,c}$ and predicted probabilities $p_{i,c}$:

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \, \ln\big(p_{i,c}\big).$$

References

[1] Ienco D, Gaetano R, Dupaquier C, Maurel P (2017) Land cover classification via multitemporal spatial data by recurrent neural networks. *arXiv preprint* arXiv:1704.04055. paper

[2] Radiant Earth Foundation (2024) South Africa Crop Type Competition. Source Cooperative repository.

[3] Capstone Group 3 (2025) Capstone_Group_3: Source code for multitemporal crop classification. *GitHub repository* Github

[4] Kordi F, Yousefi H (2022) Crop classification based on phenology information using time series of optical and synthetic-aperture radar images. *Remote Sens Appl Soc Environ* 27:100812. https://doi.org/10.1016/j.rsase.2022.100812

[5] Fan X, Chen L, Xu X, Yan C, Fan J, Li X (2023) Land Cover Classification of Remote Sensing Images Based on Hierarchical Convolutional Recurrent Neural Network. Forests 14(9):1881. https://doi.org/10.3390/f14091881

[6] Tufail R, Tassinari P, Torreggiani D (2025) Deep Learning Applications for Crop Mapping Using Multi-Temporal Sentinel-2 Data and Red-Edge Vegetation Indices: Integrating Convolutional and Recurrent Neural Networks. Remote Sens 17(18):3207. https://doi.org/10.3390/rs17183207

[7] Bandar A, Coşkunçay A (2024) Crop Classification with Attention Based BI-LSTM and Temporal Convolution Neural Network Combination for Remote Sensing BreizhCrop Time Series Data. Yüzüncü Yıl Üniversitesi Fen Bilimleri Enstitüsü Dergisi 29(1):173–188. https://doi.org/10.53433/yyufbed.1335866

[8] Mann ML, Colson L, Nealon R, Engstrom R, Nakacwa S (2023) Lite Learning: Efficient Crop Classification in Tanzania Using Traditional Machine Learning & Crowd Sourcing. *SSRN Preprint*.

[9] Saini R, Ghosh SK (2018) Crop classification on single date Sentinel-2 imagery using random forest and support vector machine. *Int Arch Photogramm Remote Sens Spatial Inf Sci* XLII-5:683–688. https://doi.org/10.5194/isprs-archives-xlii-5-683-2018

[10] Jin Z, Azzari G, You C, Di Tommaso S, Aston S, Burke M, Lobell DB (2019) Smallholder maize area and yield mapping at national scales with Google Earth Engine. *Remote Sens Environ* 228:115–128. https://doi.org/10.1016/j.rse.2019.04.016

[11] Peña-Barragán JM, Ngugi MK, Plant RE, Six J (2011) Object-based crop identification using multiple vegetation indices, textural features and crop phenology. *Remote Sens Environ* 115:1301–1316. https://doi.org/10.1016/J.RSE.2011.01.009

[12] Belgiu M, Drăguţ L (2016) Random forest in remote sensing: A review of applications and future directions. *ISPRS J Photogramm Remote Sens* 114:24–31. https://doi.org/10.1016/j.isprsjprs.2016.01.011

[13] Rangarajan AK, Purushothaman R, Prabhakar M, Szczepański C (2023) Crop identification and disease classification using traditional machine learning and deep learning approaches. *J Eng Res* 11(1):228–252. https://doi.org/10.36909/jer.11941

[14] Gadiraju KK, Ramachandra B, Chen Z, Vatsavai RR (2020) Multimodal Deep Learning Based Crop Classification Using Multispectral and Multitemporal Satellite Imagery. In: *Proc 26th ACM SIGKDD Int Conf Knowl Discov Data Min*. https://doi.org/10.1145/3394486.3403375

[15] Rußwurm M, Körner M (2018) Multi-Temporal Land Cover Classification with Sequential Recurrent Encoders. *ISPRS Int J Geo-Inf* 7(4):129. https://doi.org/10.3390/ijgi7040129

[16] Radiant Earth (2024) Spot The Crop Challenge. GitHub repository.

[17] Campbell JB, Wynne RH (2011) *Introduction to Remote Sensing*. Guilford Press.

[18] Wardlow BD, Egbert SL, Kastens JH (2007) Analysis of time-series MODIS 250 m vegetation index data for crop classification in the U.S. Central Great Plains. *Remote Sens Environ* 108(3):290–310. https://doi.org/10.1016/j.rse.2006.11.021

[19] James G, Witten D, Hastie T, Tibshirani R (2013) *An Introduction to Statistical Learning*. Springer.

[20] Breiman L (2001) Random forests. *Machine Learning* 45(1):5–32. https://doi.org/10.1023/A:1010933404324

[21] Ke G, Meng Q, Finley T, et al. (2017) LightGBM: A highly efficient gradient boosting decision tree. *NeurIPS* 30.

[22] Chen T, Guestrin C (2016) XGBoost: A scalable tree boosting system. In: *Proc 22nd ACM SIGKDD Int Conf Knowl Discov Data Min*, pp 785–794. https://doi.org/10.1145/2939672.2939785

[23] Friedman JH (2001) Greedy function approximation: A gradient boosting machine. *Ann Stat* 29(5):1189–1232. https://doi.org/10.1214/aos/1013203451

[24] Maxwell AE, Warner TA, Fang F (2018) Implementation of machine-learning classification in remote sensing: an applied review. *Int J Remote Sens* 39(9):2784–2817. https://doi.org/10.1080/01431161.2018.1433343

[25] Zhu XX, Tuia D, Mou L, et al. (2017) Deep learning in remote sensing: A comprehensive review and list of resources. *IEEE Geosci Remote Sens Mag* 5(4):8–36. https://doi.org/10.1109/MGRS.2017.2762307

[26] Kussul N, Lavreniuk M, Skakun S, Shelestov A (2017) Deep Learning Classification of Land Cover and Crop Types Using Remote Sensing Data. *IEEE Geosci Remote Sens Lett* 14(5):778–782. https://doi.org/10.1109/LGRS.2017.2681128

[27] Muruganantham P, Wibowo S, Grandhi S, Islam N (2024) A Deep Learning Approach for Dealing with Tabular Data in Crop Classification. *Proc IEEE Int Conf ICT Convergence (ICTC)*:2054–2059. doi:10.1109/ICTC62082.2024.10826760

[28] Arik SÖ, Pfister T (2021) TabNet: Attentive Interpretable Tabular Learning. *Proc AAAI Conf Artif Intell* 35(8):6679–6687. https://doi.org/10.1609/aaai.v35i8.16826

[29] Garnot VSF, Landrieu L (2020) Lightweight Temporal Self-attention for Classifying Satellite Images Time Series. In: *Advanced Analytics and Learning on Temporal Data (AALTD 2020), ECML PKDD Workshops*, LNCS vol 12588, pp 171–181. Springer. https://doi.org/10.1007/978-3-030-65742-0_12

[30] Pelletier C, Webb GI, Petitjean F (2019) Temporal Convolutional Neural Network for the Classification of Satellite Image Time Series. *Remote Sens* 11(5):523. https://doi.org/10.3390/rs11050523

[31] Zhong L, Hu L, Zhou H (2018) Deep learning based multi-temporal crop classification. *Remote Sens Environ* 221:430–443. doi:10.1016/j.rse.2018.11.032

[32] Rustowicz RM, Cheong R, Wang L, Ermon S, Burke M, Lobell D (2019) Semantic Segmentation of Crop Type in Africa: A Novel Dataset and Analysis of Deep Learning Methods. In: *Proc IEEE/CVF CVPR Workshops*, pp 75–82.

[33] Wang H, Chang W, Yao Y, Yao Z, Zhao Y, Li S, Liu Z, Zhang X (2023) Cropformer: A new generalized deep learning classification approach for multi-scenario crop classification. *Front Plant Sci* 14:1130659. https://doi.org/10.3389/fpls.2023.1130659

[34] Mahlayeye M, Darvishzadeh R, Nelson A (2024) Characterising maize and intercropped maize spectral signatures for cropping pattern classification. *Int J Appl Earth Obs Geoinf* 128:103699. https://doi.org/10.1016/j.jag.2024.103699

[35] Mann ML, Warner JM (2017) Ethiopian wheat yield and yield gap estimation: A spatially explicit small area integrated data approach. *Field Crops Res* 201:60–74. https://doi.org/10.1016/j.fcr.2016.10.014

[36] Mann ML, Warner JM, Malik AS (2019) Predicting high-magnitude, low-frequency crop losses using machine learning: an application to cereal crops in Ethiopia. *Clim Change* 154:211–227. https://doi.org/10.1007/s10584-019-02432-7

[37] Mann ML (2025) xr_fresh: Automated Time Series Feature Extraction for Remote Sensing and Gridded Data. *J Open Source Softw* 10(115):9009. https://doi.org/10.21105/joss.09009

[38] Teixeira I, Morais R, Sousa JJ, Cunha A (2023) Deep Learning Models for the Classification of Crops in Aerial Imagery: A Review. *Agriculture* 13(5):965. https://doi.org/10.3390/agriculture13050965

[39] Höhl A, Obadic I, Fernández-Torres MÁ, Oliveira D, Zhu XX (2024) Recent Trends Challenges and Limitations of Explainable AI in Remote Sensing. In: *Proc IEEE/CVF Conf Comput Vis Pattern Recognit*, pp 8199–8205.

[40] Buttar PK et al. (2024) Satellite imagery analysis for crop type segmentation using U-net architecture. *Procedia Comput Sci* 235:3418–3427. https://doi.org/10.1016/j.procs.2024.04.322

[41] Liu N, Zhao Q, Williams R, Barrett B (2024) Enhanced crop classification through integrated optical and SAR data: a deep learning approach for multi-source image fusion. *Int J Remote Sens* 45(19–20):7605–7633. https://doi.org/10.1080/01431161.2023.2232552

[42] Pankajakshan P, Padmasanan A, Sundar S (2025) Deep Architectures Fail to Generalize: A Lightweight Alternative for Agricultural Domain Transfer in Hyperspectral Images. *Sensors* 26(1):174. https://doi.org/10.3390/s26010174

[43] Espejo-Garcia B, Mylonas N, Athanasakos L, Fountas S, Vasilakoglou I (2020) Towards weeds identification assistance through transfer learning. *Comput Electron Agric* 171:105306. https://doi.org/10.1016/j.compag.2020.105306

[44] Koklu M, Unlersen MF, Ozkan IA, Aslan MF, Sabanci K (2022) A CNN-SVM study based on selected deep features for grapevine leaves classification. *Measurement* 188:110425. https://doi.org/10.1016/j.measurement.2021.110425

[45] Yang S, Gu L, Li X, Jiang T, Ren R (2020) Crop Classification Method Based on Optimal Feature Selection and Hybrid CNN-RF Networks for Multi-Temporal Remote Sensing Imagery. *Remote Sens* 12(19):3119. https://doi.org/10.3390/rs12193119

[46] Nowakowski A, Mrziglod J, Spiller D, Bonifacio R, Ferrari I, Mathieu PP, Garcia-Herranz M, Kim DH (2021) Crop type mapping by using transfer learning. *Int J Appl Earth Obs Geoinf* 98:102313. https://doi.org/10.1016/j.jag.2021.102313

[47] Bégué, A., Leroux, L., Soumaré, M., Faure, J.F., Diouf, A.A., Augusseau, X., Touré, L., Tonneau, J.P., 2020. Remote sensing products and services in support of agricultural public policies in Africa: overview and challenges. *Frontiers in Sustainable Food Systems*, 4. https://doi.org/10.3389/fsufs.2020.00058